The Fresnel Stage

How Stage 1's Fresnel magnifier and combiner turn an 80 × 40 mm patch of an LCD into a picture floating 4.9 m ahead of the driver, traced through the real lens, groove by groove.

2026-10-25 · Companion to `docs/ARHUD-PLAN-TFT.pdf`, section 5 (Stage 1) · Numbers from `docs/arhud-fresnel-src/calc.py` and its ray tracer; renders from Blender Cycles

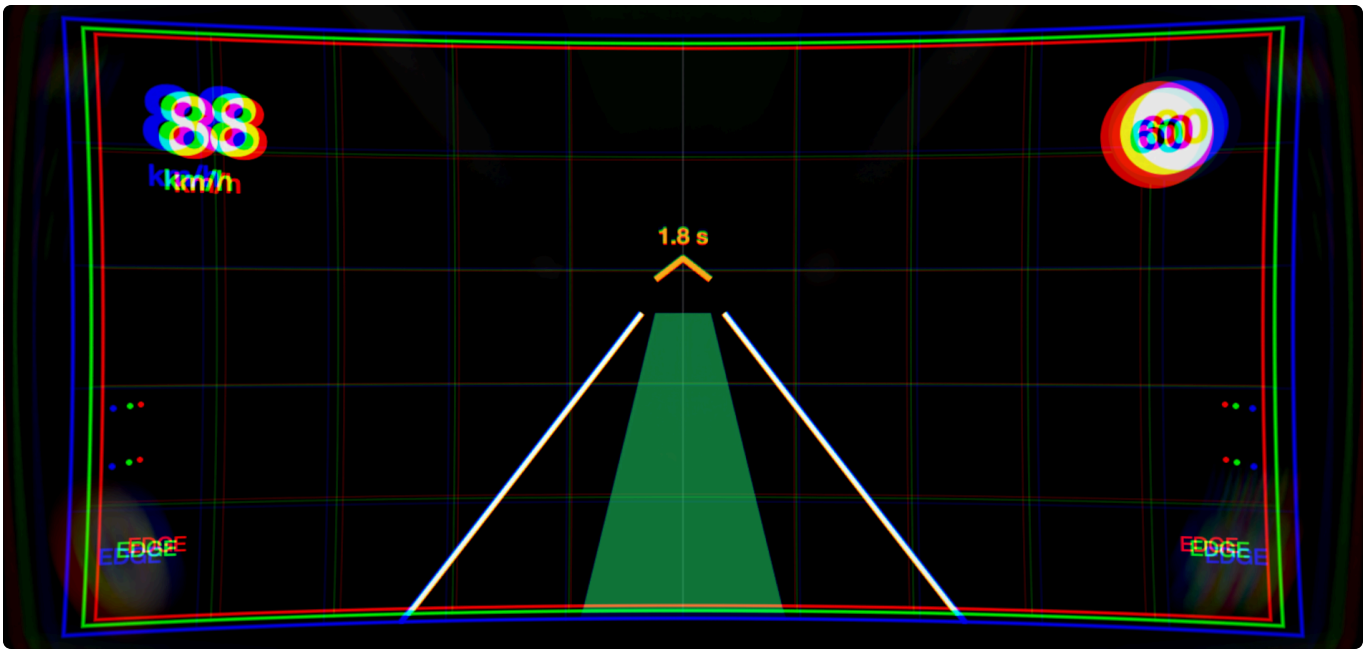


Figure 1. What the eye sees: a Cycles render through a 300 mm PMMA Fresnel lens built tooth by tooth (0.5 mm pitch), with a 4 mm pupil focused at the virtual image and the lens's dispersion rendered as three colour passes. The panel shows a HUD test frame on the 80 × 40 mm crop; the combiner is left out (it is a flat mirror). Section 5 explains every artefact in it.

AT A GLANCE

Stage 1 is a magnifying glass. A 300 mm plastic Fresnel lens sits 280 mm above the panel, just inside its 300 mm focal length, so the eye sees an enlarged, upright image of the panel far behind it. A tilted sheet of glass, the combiner, reflects that image toward the driver while letting the road show through.

4.9 m	**14.7° × 7.1°**	**141 × 251 mm**
from the eye to the virtual image at the centre (4.2 m from the lens)	field of view from the eye, traced (the plan's 16° × 8° is measured from the lens)	eye box set by the lens aperture, traced at 700 mm (ASSUMED eye distance)
225 mm/mm	**86%**	**39 W**
image distance change per mm of panel movement, but only 0.0128 D: invisible to focus	of the eye's sight lines at the field edge pass a facet cleanly; the rest strike draft walls	of sunlight the lens can focus onto the panel, peaking at 1,528 kW/m² 12° off axis

What this document adds to the plan

1. **Quote the image from the eye.** The image is 4.9 m from the eye, not 4.2 m, and spans 14.7° × 7.1° at 60 px per degree on the 4K monitor (86 on the 5.5-inch panel). The plan's figures are measured from the lens (section 3).
2. **The image is not flat.** It curves away from the eye: 4.9 m at the centre, 5.5–7.6 m at the side edges. The eye cannot tell (0.06 D), but registration should use a measured distance per point, as Stage 2 does (section 4).
3. **Set the panel distance per lens, by parallax.** A ±2% focal-length tolerance moves the image between 3.3 and 5.9 m. Put the panel on a slide and set it against a target (section 11).
4. **Grooves must face the eye.** Flipped, only 73% of the light at r = 100 mm passes cleanly (against 98%), and the field edge shows 0.85 D of astigmatism (section 7).
5. **Per-colour pre-distortion.** Red and blue separate by 0.36° at the field edge (31 pixels on the 5.5-inch panel); the split moves with the head (section 6).
6. **Stage 1 needs its own sun defence.** It has no cold mirror, filter or shutter. Sunlight straight down the lens puts 121 kW/m² on the panel, and 1,528 kW/m² at 12° off axis. A lid that is closed whenever the HUD is off is the minimum (section 10).
7. **The path must be folded.** 280 mm does not fit straight on a dash top; a 45° front-surface mirror makes a box about 410 × 116 mm (section 11; a suggestion, not in the plan).
8. **Crop and backlight disagree.** Stage 1 uses an 80 × 40 mm crop; the custom backlight is designed for 60 × 30 mm. Either light 1.78× the area or accept 10.8° × 5.3° in Stage 1.

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Values marked ASSUMED are not in the plan or a datasheet: the eye distance (700 mm along the light path), the in-car layout, the windshield and combiner transmissions, and the lens's design conjugate. The lens is the plan's Knight Optical LFQ300300: 300 × 300 mm, f = 300 mm, 2.0 mm thick, 0.5 mm groove pitch, PMMA (stock listing).

1 Stage 1 at a glance

The plan builds the HUD's optics in three stages. Stage 0 is a desk rig for the software. Stage 2 is the real design: a concave mirror that bounces the panel's light off the windshield. Stage 1 sits between them. It puts a working HUD in the car within weeks, before the custom mirror exists, so registration can be tested on real roads at night and in overcast weather. It is also the demo fallback if the mirror is late.

Stage 1 has four parts, from the bottom of the light path to the top:

- **The panel**, an LCD showing the HUD graphics. Only an 80 × 40 mm patch of it, the *crop*, is used: the plan's 16-inch 4K monitor first, the 5.5-inch 1080p panel later.
- **The Fresnel lens**, a 300 × 300 mm sheet of clear acrylic (PMMA), 2.0 mm thick, with fine circular grooves on one face. It acts as a strong magnifying glass with a 300 mm focal length. It sits 280 mm from the panel.
- **The combiner**, a sheet of glass that is partly reflective (70% of light passes, 30% reflects). It stands above the dash, tilted so that it reflects the lens's light toward the driver while the road shows through it.
- **The eye.** Looking through the combiner, the driver sees the graphics floating over the road about 4.9 m ahead.

Figure 2 shows the arrangement in a car, to scale, with three rays traced through a model of the real lens. Two things stand out. The image is far beyond the windshield, near the road, which is what lets graphics line up with real objects. And the 280 mm between the panel and the lens is long: drawn straight down from a lens sitting on the dash, the panel would end up 178 mm inside the dashboard. Section 11 folds the path with a mirror.

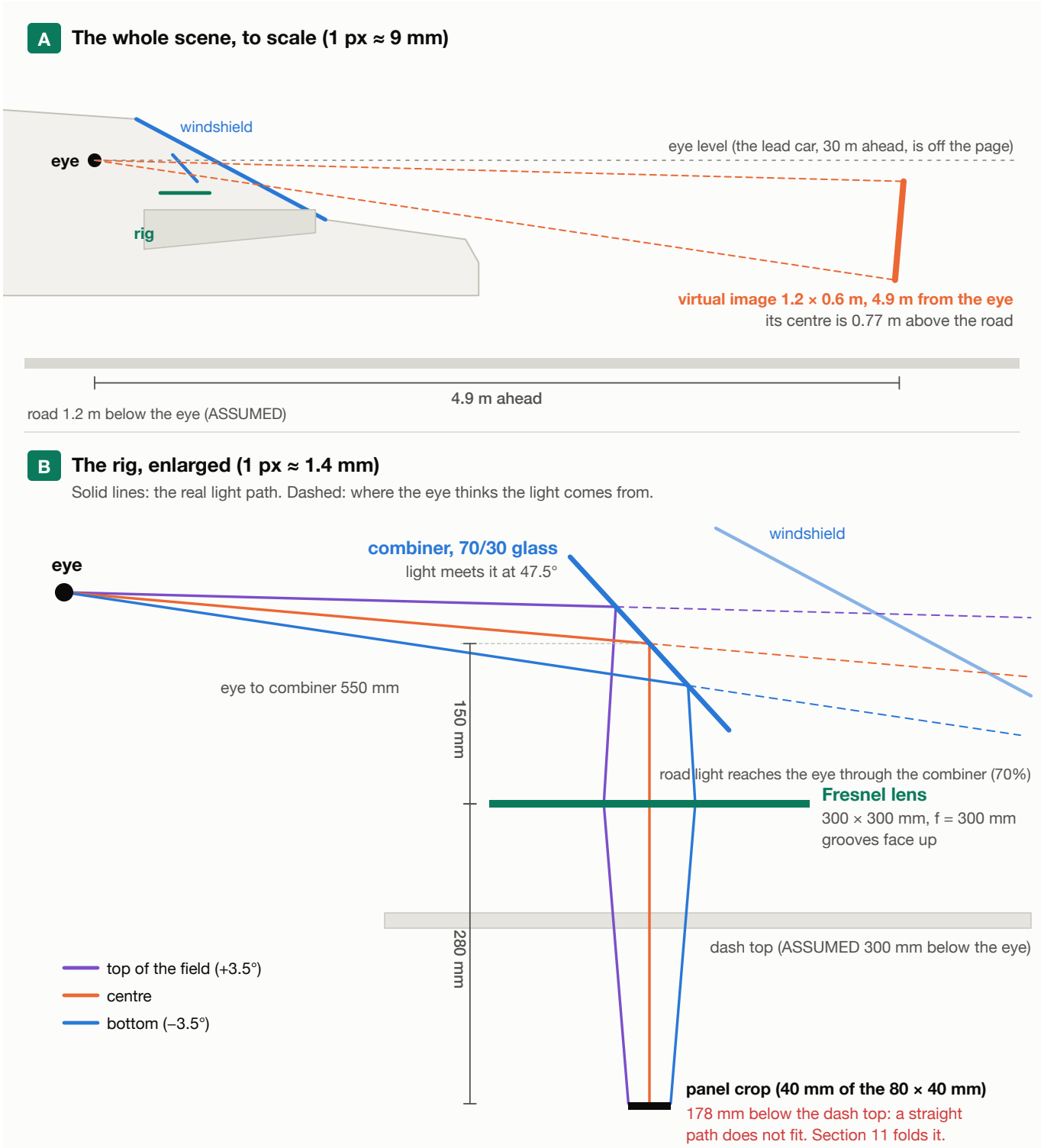


Figure 2. Stage 1 in the car. A: the whole scene to scale, with the virtual image 4.9 m from the eye. B: the rig enlarged, with rays to the top, centre and bottom of the field traced through the lens model. The car outline, dash height and the split of the eye-to-lens distance are ASSUMED for illustration.

Straightening the light path

The combiner is flat, and a flat mirror changes the direction of light without changing any distance. So for working out the optics, the bend at the combiner can be removed: picture the eye looking straight down a single line through the lens at the panel (Figure 3). Everything in sections 2 to 8 uses this unfolded view. The eye is 700 mm from the lens along the light path (550 mm to the combiner, 150 mm down to the lens). The plan does not fix this distance; it is an assumption, and Table 1 shows what changes if it is 600 or 800 mm instead.

The combiner is a flat mirror, so the path can be straightened onto one axis.

Distances along the axis to scale (0.5 px per mm); heights ×3 so the rays can be told apart.

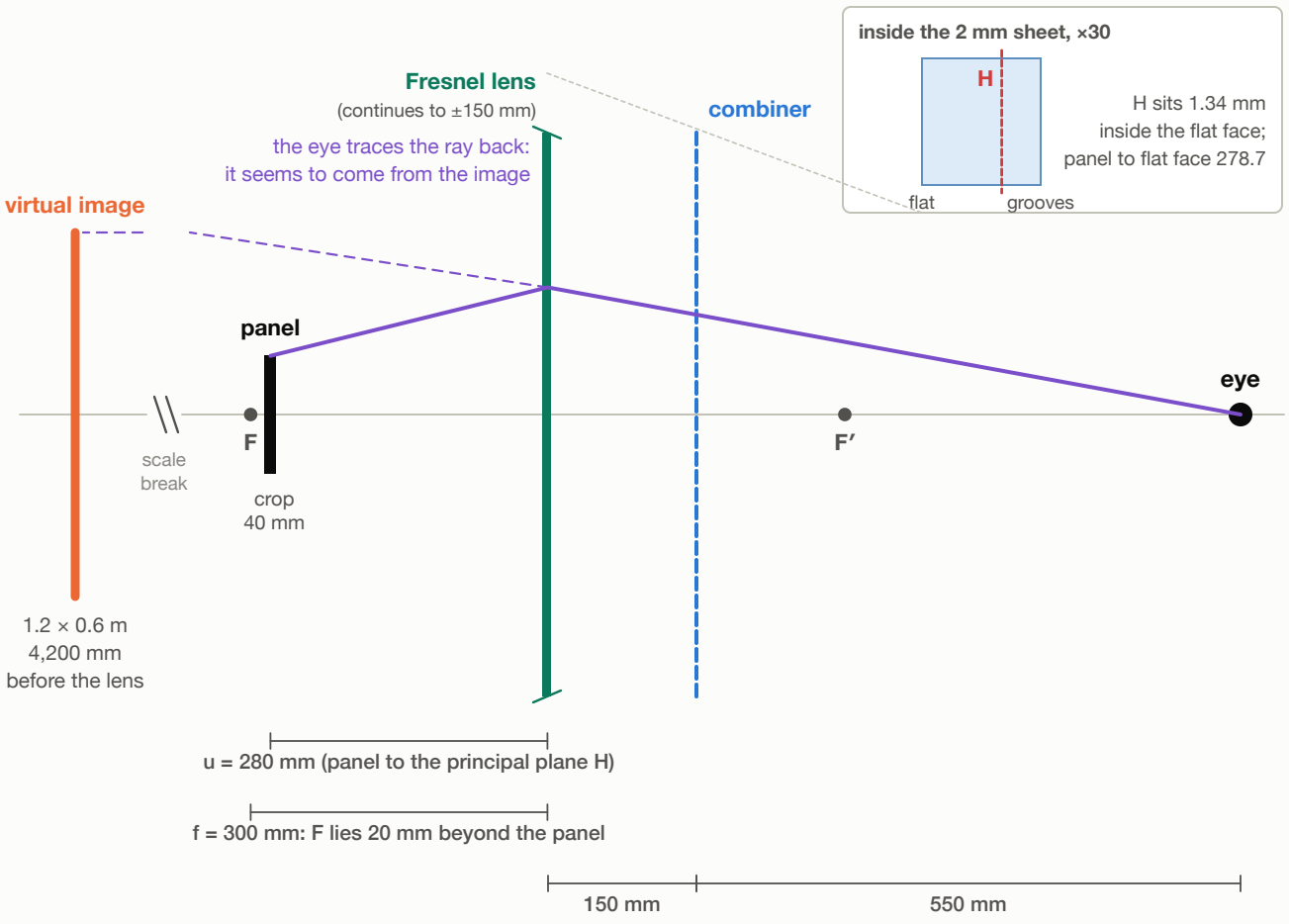


Image distance from the eye: 4,200 + 700 mm = 4.9 m. Field of view from the eye: 14.7° × 7.1°.

The plan's 16° × 8° is the same image measured from the lens, not the eye: 2 × atan(0.6 ÷ 4.2).

Figure 3. The light path unfolded. Along the axis the drawing is to scale except for the break before the virtual image; heights are tripled. The inset shows where the lens's principal plane H sits inside the 2 mm sheet; the 280 mm is measured from it.

Two details in Figure 3 matter later. The focal point F lies only 20 mm beyond the panel, which makes the image distance sensitive to the panel's position (section 4) and makes the lens a dangerous sun concentrator (section 10). And the distance "280 mm" has to be measured from somewhere: for a thick lens

the reference is its principal plane H, which for this sheet sits 1.34 mm inside the flat face. Measured from the flat face, the panel is 278.7 mm away.

EYE TO LENS (LIGHT PATH)	FIELD FROM THE EYE	IMAGE FROM THE EYE	PX/DEG, 4K MONITOR	EYE BOX, SIMPLE ESTIMATE
600 mm	14.9° × 7.2°	4.8 m	58	143 × 224 mm
700 mm (used here, ASSUMED)	**14.7° × 7.1°**	**4.9 m**	**60**	**120 × 213 mm**
800 mm	14.4° × 7.0°	5.0 m	61	98 × 203 mm

Table 1. What the assumed eye distance changes, traced through the lens model. The simple eye box is the lens half-width minus the edge ray's offset; the traced value at 700 mm, which lets the edge angle change as the head moves, is 141 × 251 mm (section 5).

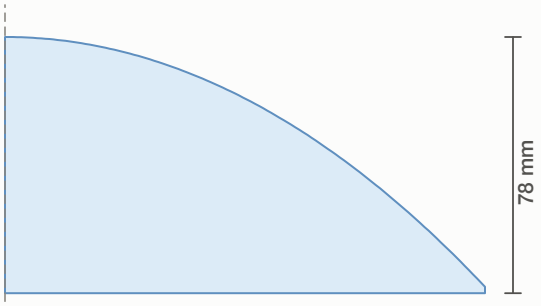
2 What a Fresnel lens is

An ordinary lens bends light with its curved surfaces: where the surface is tilted, light crossing it changes direction. All the bending happens at the surface; the glass inside only carries the light through. A Fresnel lens keeps the surface and drops most of the inside.

Figure 4 shows the idea with this lens's real numbers. A solid lens with a 300 mm focal length and a 300 mm diameter would be a dome 78 mm thick, weighing about 3.3 kg in PMMA. (A simple spherical surface cannot even reach 300 mm: its radius would be 147 mm, so it closes into a hemisphere at 294 mm. The surface has to be aspheric.) Cut that dome into thin rings, keep only the slanted top of each ring, and lower every piece onto a thin base. What remains is a 2.0 mm sheet weighing 212 g with 300 concentric teeth from the centre to the edge, each with the same slope the solid lens had at that radius.

A **A solid lens that does the same job**

Same focal length (300 mm), round, 300 mm across, drawn to scale (centre to edge). Thickness at the centre: 78 mm plus the 2 mm base.

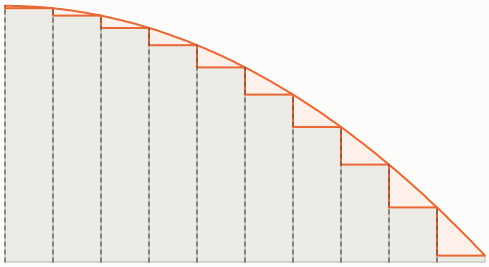


weight ≈ 3.3 kg of PMMA

(a 300 mm square would be heavier: its corners need 159 mm of sag)

A spherical lens can't do it at all: R = 147 mm, so its widest possible size is 294 mm.

B **Cut it into rings (15 mm wide here so you can see them; the real ones are 0.5 mm)**



orange: the thin sliver of curved surface on each ring.

Grey: material underneath that only carries light straight through. It adds no bending, so it can go.

C **Drop every sliver onto a thin base: the result is a Fresnel lens**



Each tooth's slanted face has the same angle as the solid lens at that radius, so light bends the same way. The vertical steps do no useful work.

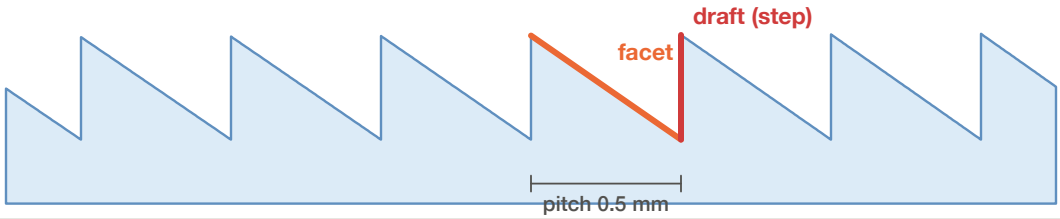
D **The real lens: a 2.0 mm sheet, 300 teeth from centre to edge (424 to the corner), 212 g**



at this scale the teeth are a fuzz on the top surface

Six teeth around r = 100 mm, magnified 150×. Each is 0.5 mm wide and 0.35 mm tall here.

left: toward the lens centre; right: toward the edge



the sheet continues below (2 mm in all)

Figure 4. From a solid lens to a Fresnel lens, drawn from the profile solved for this lens. A to C use 15 mm rings so the steps are visible; D shows the real 0.5 mm teeth at 150× near r = 100 mm.

The parts of a tooth

Each tooth has two faces (Figure 5). The **facet** is the slanted face that does the work: its angle to the sheet equals the slope of the solid lens at that radius. The **draft wall** is the near-vertical step back up to the next tooth. It is there only because the surface has to return to the base; light that hits it goes somewhere it should not, which is the source of most Fresnel artefacts (section 8). The spacing of the teeth is the **pitch**, 0.5 mm on the plan's lens, and it is the same everywhere, so the teeth get deeper toward the edge, where the facets are steeper.

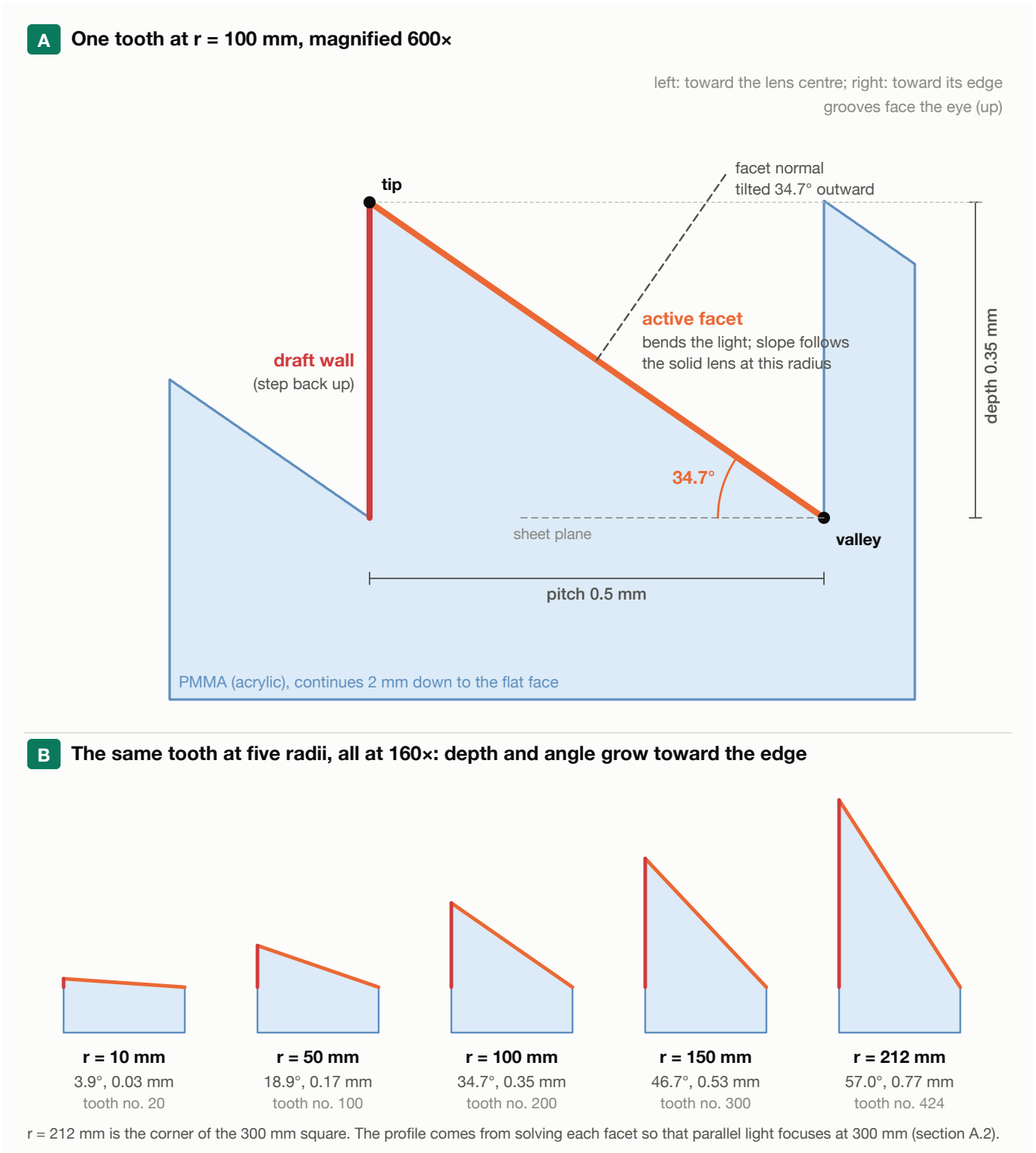
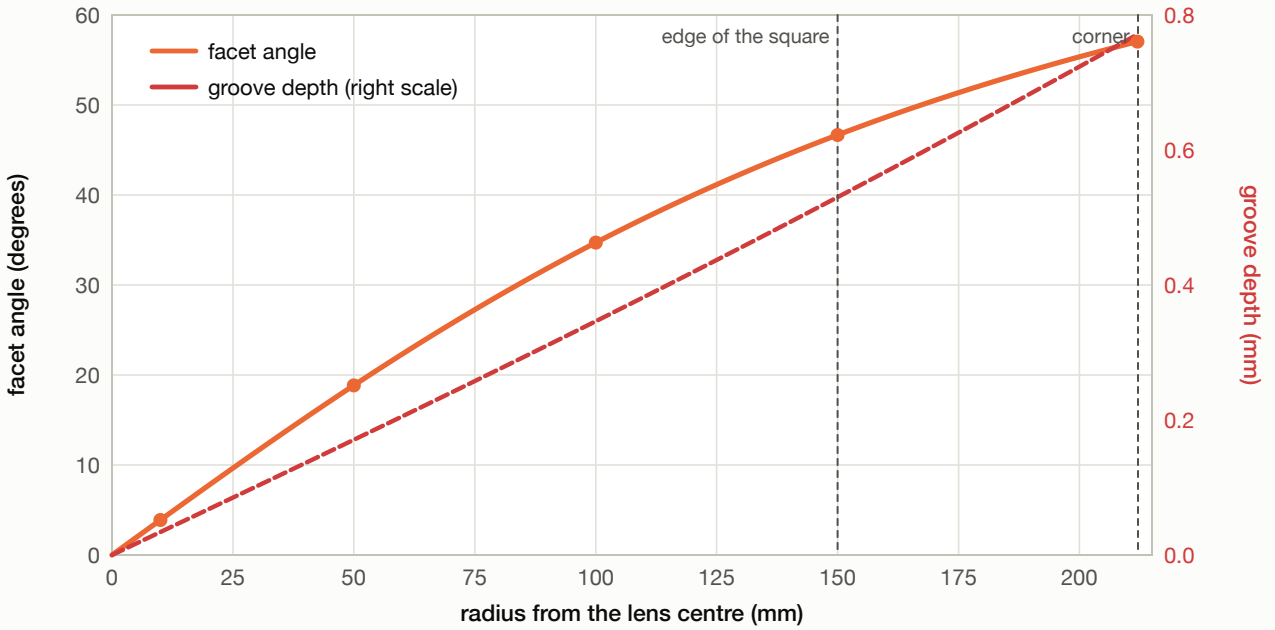


Figure 5. The teeth of this lens. A: one tooth at r = 100 mm with every part named. B: the tooth at five radii, from almost flat near the centre to 57° at the corner of the 300 mm square.



Pitch is constant (0.5 mm), so the depth is pitch × tan(angle). Critical angle in PMMA: 42.1°.

Figure 6. Facet angle and groove depth against radius. The facet angle rises from 0° at the centre to 46.7° at the edge of the square and 57.0° at its corners.

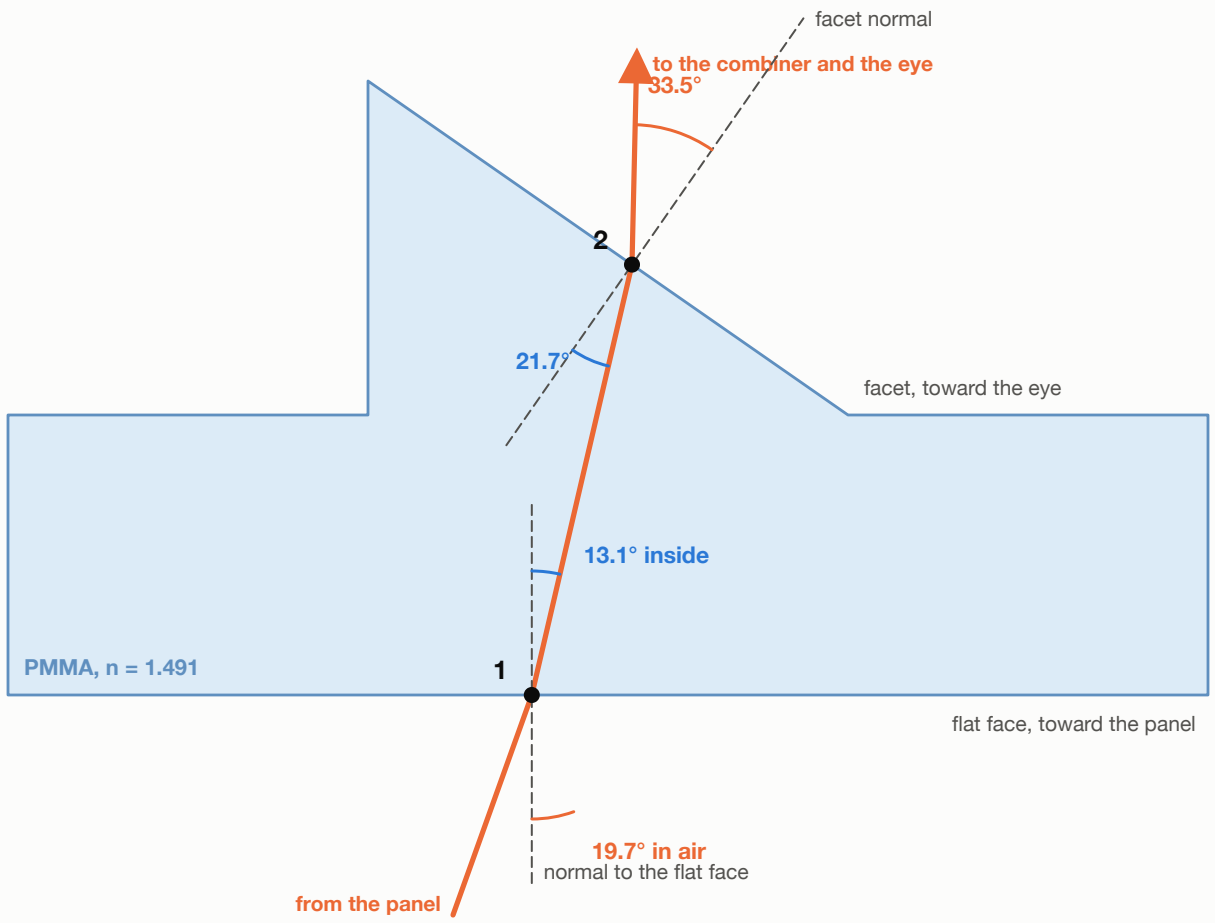
The facet angles come from a calculation, not a catalogue: the manufacturer does not publish the profile. Stock Fresnel lenses are designed so that a parallel beam arriving on the grooved side comes to a sharp focus on the flat side (an ASSUMED design, standard for stock lenses), and solving that condition ray by ray gives a unique facet angle at each radius (Appendix A.2). Every number and figure in this document uses that profile.

One ray, followed through

Figure 7 follows a single ray from the centre of the panel through the lens at r = 100 mm. Two refractions do the bending. Entering the flat face, the ray slows down in the plastic and bends toward the perpendicular; leaving through the facet, it speeds up again and bends away from the facet's perpendicular, which is tilted 34.8°. The rule at each surface is Snell's law, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, with n = 1 in air and 1.491 in PMMA. The ray arrives at 19.7° from the axis and leaves at 1.33°: almost parallel to the axis, and spreading slightly outward, exactly as if it had come from a point 4.2 m behind the lens. That is the virtual image of section 3.

A ray from the centre of the panel to r = 100 mm, followed through both surfaces

Angles are the traced values. The tooth and the sheet are drawn large; the angles are true.



1 Flat face.	The ray leaves the panel at 19.7° from the axis, slows in the plastic and bends toward the normal: **sin 19.7° = 1.491 × sin 13.1°**
2 Facet.	Inside, it meets the facet at 21.7° to the facet's normal (tilted 34.8° outward), then bends away from the normal: **1.491 × sin 21.7° = sin 33.5°, so it leaves at 34.8° – 33.5° = 1.33° from the axis**
Result.	19.7° becomes 1.33°: almost parallel, slightly outward, as if it came from a point 4.2 m behind the lens (100 mm ÷ 4,200 mm ≈ 1.4°). Reflection at the two surfaces costs 7.9%.
Trapping.	21.7° is far below PMMA's 42.1° critical angle, so no light is trapped by total internal reflection here.

Figure 7. One ray through the sheet at r = 100 mm, with the traced angles. The sheet and tooth are drawn large; the angles are true.

Why PMMA

Nearly every large Fresnel lens is moulded or cast in PMMA (acrylic), including the two lenses the plan lists. It transmits about 92% of visible light (two surfaces), it bends colours apart only a little more than optical glass (Abbe number 58 against 64 for BK7 glass and 28 for polycarbonate, section 6), and it has low internal

stress, so it barely disturbs the LCD's polarised light (section 9). Its weaknesses are heat and scratches. PMMA softens near 96 °C and is rated for continuous use at 65–95 °C, below what a dashboard reaches in a parked car in summer, so the lens should not be left on the dash in the sun (section 10 adds a second reason). It also expands about ten times as much as glass with temperature, which moves the image slightly (section 4). Clean it with water and a soft cloth; ammonia and alcohol cleaners can craze it.

3 How the lens makes a virtual image

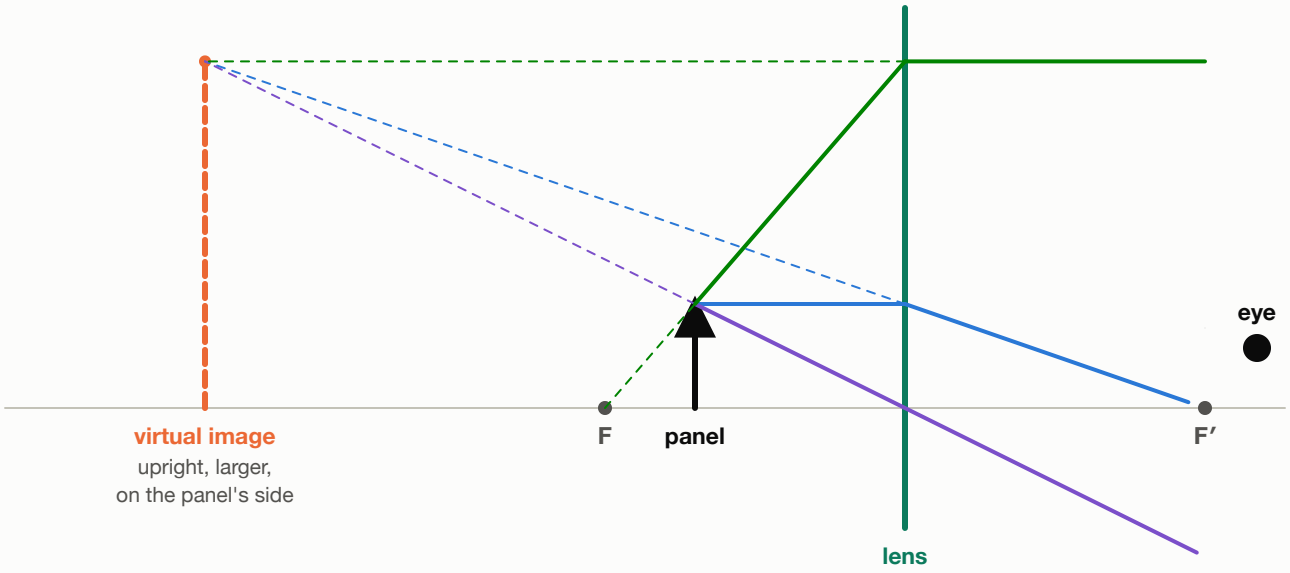
Hold a magnifying glass close to a page and the print looks larger and farther away. Stage 1 is the same trick at a larger scale: the panel sits just inside the lens's focal length, so the lens cannot bring its light to a focus. Instead, the light leaves the lens spreading out, as if it came from a large picture far behind the panel. The eye cannot tell the difference between light that really came from there and light that only appears to, so it sees the large picture. Optics calls it a *virtual image*: nothing is actually at that place, and a screen held there would show nothing.

Figure 8 constructs the image with three easy rays. A ray leaving the panel parallel to the axis is sent through the far focal point F′. A ray through the centre of the lens goes straight on. A ray that starts as if from the near focal point F leaves parallel to the axis. After the lens the three rays spread apart and never meet; traced backward, they meet at one point behind the panel, and every other ray from the same panel point appears to come from that same point.

A **The construction, with the proportions changed so it fits (u = 0.7 f here)**

- 1 leaves the panel parallel to the axis, so the lens sends it through F′
- 2 passes through the lens centre without bending
- 3 starts as if from F, so the lens sends it out parallel to the axis

The three rays spread apart after the lens: they never meet on the eye's side, so no real image forms there.
Traced backward (dashed) they meet at one point behind the panel. That point is what the eye sees: a virtual image.



B **The same construction with Stage 1's numbers**

Lens equation	1/v = 1/f – 1/u = 1/300 – 1/280 so v = –4,200 mm (the minus sign means: on the panel's side, virtual)
Magnification	M = |v| / u = 4,200 / 280 = 15: the 80 × 40 mm crop becomes 1.2 × 0.6 m
Why so large	u is 20 mm short of f. As u approaches f, f – u shrinks and both |v| = uf / (f – u) and M = f / (f – u) grow without limit.
From the eye	the image is 4,200 + 700 = 4.9 m away and fills 14.7° × 7.1° (traced, including distortion).
Pixels	80 mm is 872 px on the 4K monitor (60 px/deg) or 1,261 px on the 5.5-inch panel (86 px/deg).

Figure 8. The virtual image constructed from three rays (A), and the same construction with Stage 1's numbers (B). In A the panel is drawn at 0.7 f instead of 0.93 f, or the image would be off the page.

With the plan's numbers, the lens equation 1/v = 1/f – 1/u gives v = –4,200 mm: the image is 4.2 m from the lens, on the panel's side (the minus sign). It is 15 times larger than the panel, so the 80 × 40 mm crop becomes 1.2 × 0.6 m. These figures are measured from the lens. The driver's eye is another 700 mm back, so the image is 4.9 m from the eye, and its angular size, traced through the lens model with its distortion, is 14.7° × 7.1° rather than the plan's 16° × 8°. Seen from the eye, the 4K monitor's 872 pixels across the crop give 60 pixels per degree (1.01′ each, at the eye's limit of about 1′), and the 5.5-inch panel's 1,261 give 86 (0.70′ each).

Why the image is so far away and so large

Both $|v| = uf / (f - u)$ and $M = f / (f - u)$ have $f - u$ underneath. The panel is only 20 mm short of the focal length, so $f - u$ is small and both are large. Moving the panel toward the focal point sends the image off toward infinity; moving it toward the lens brings the image in and shrinks it. Section 4 is about how sensitive that is.

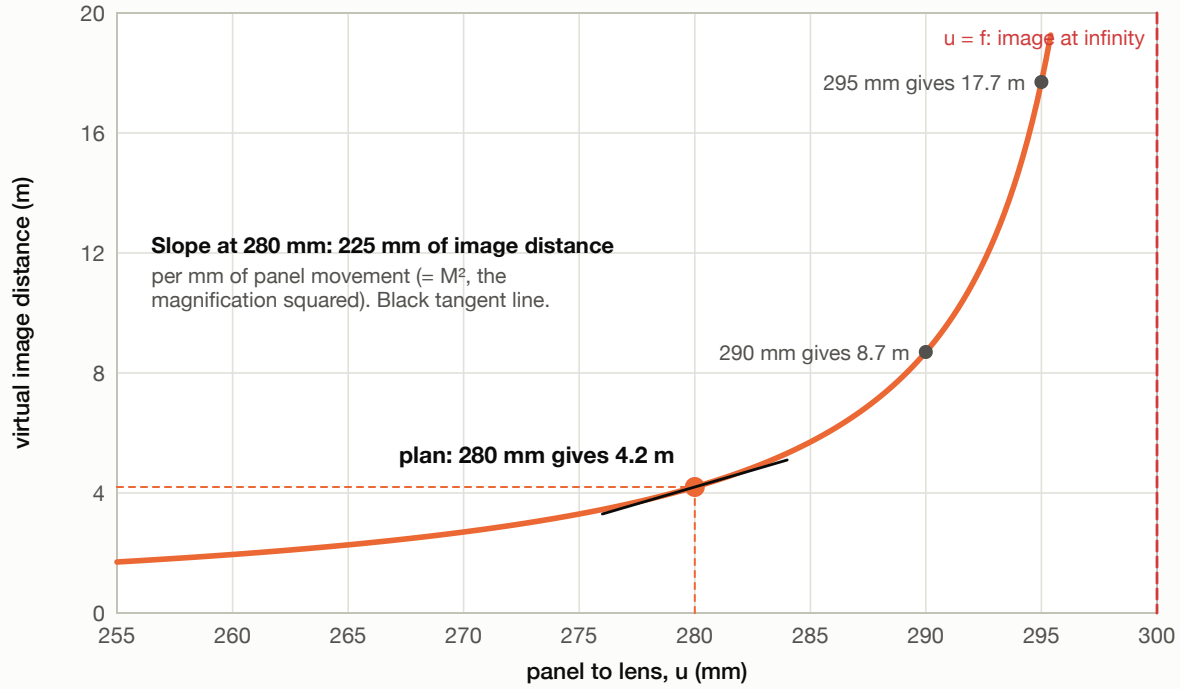
A magnifier does not change brightness. The image is larger, but the light from each panel point is spread over a proportionally larger image area, and the eye receives the same luminance it would get from the panel itself, minus the lens's losses (section 9). That is why the HUD's brightness has to come from the panel and its backlight, not from the optics.

4 Where the image sits, and how stable it is

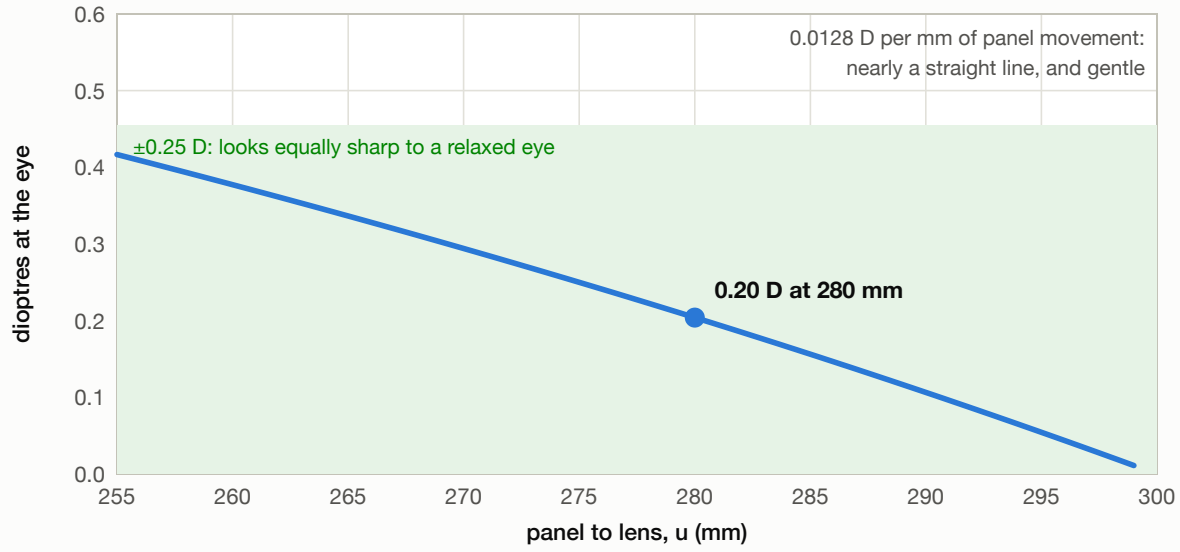
Because the panel sits so close to the focal point, small changes move the image a long way. Figure 9A plots the image distance against the panel distance. At 280 mm the slope is 225 mm per mm: moving the panel 1 mm toward the focal point pushes the image about 0.2 m farther away, and with the panel at 295 mm the image is already 17.7 m away. In metres this looks alarming.

The eye, though, does not focus in metres. It focuses in *dioptres*, one over the distance in metres, and on that scale the curve is gentle and nearly straight (Figure 9B): 0.0128 D per mm of panel movement. A relaxed eye sees everything within about ±0.25 D of its focus equally sharply, which here covers panel positions from about 260 mm to beyond 295 mm. So the exact image distance hardly matters for how sharp the HUD looks. What it does matter for is *parallax*, explained below.

A **Image distance from the lens, for panel distances 255 to 299 mm**



B **The same curve in dioptres (1 ÷ distance from the eye in metres): the scale the eye focuses on**



In metres the image distance looks wildly sensitive; in dioptres it is not. The eye cannot tell 4 m from 6 m by focus.

What the distance does change is parallax: where a symbol appears against the road when the head moves (Figure 11).

Figure 9. Image distance from the lens against panel distance (A), and the same in dioptres at the eye (B), from the thin-lens equation (the traced lens agrees on axis to 0.1 mm).

The lens's tolerance, and its temperature

Two things move the image even when the panel stays put. The first is the lens itself: the focal length of a stock Fresnel lens is only specified to ±2% for the precision grade and ±5% for the standard grade (Knight Optical). A precision lens could therefore put the image anywhere from 3.3 m to 5.9 m; a standard one from 2.5 m to 16.0 m (Figure 10A). The fix is simple: treat 280 mm as a starting point, mount the panel on a slide, and set its distance for the lens in hand (section 11). For the precision grade the slide needs about 274.8–285.2 mm of travel.

The second is temperature. Warming PMMA makes it expand (70 parts per million per kelvin) and lowers its refractive index (about -1.05×10^{-4} per kelvin); both lengthen the focal length. A lens 40 K warmer than when it was set has f = 303.4 mm and puts the image at 3.63 m instead of 4.2 m (Figure 10B). A glass lens would hardly move. In practice the cabin at night is fairly stable, but the lens sits right above a 12 W backlight in Stage 1's box, so set the distance after the rig has warmed up, and keep the box ventilated.

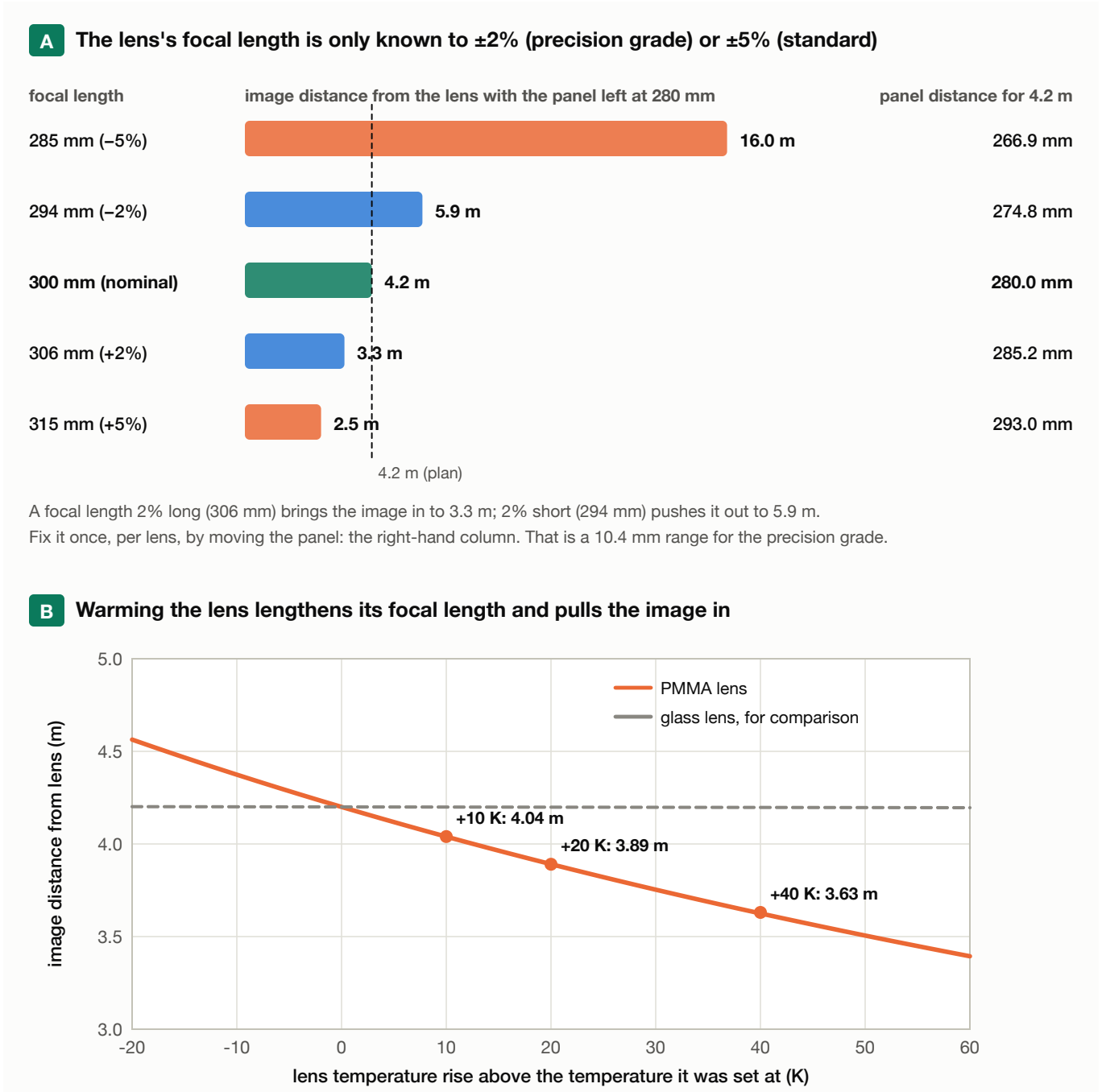


Figure 10. A: image distance for the lens's focal-length tolerance, and the panel distance that restores 4.2 m. B: image distance against the lens's temperature rise, PMMA against glass (coefficients approximate).

Why the distance matters: parallax

A HUD symbol is drawn on the virtual image, about 5 m away; the car it marks is perhaps 30 m away. Seen from one head position they line up. Move the head sideways and they separate, the way a finger held up at arm's length jumps against the background when you switch eyes (Figure 11). The separation is the head

movement times ($1/D_{image} - 1/D_{target}$): 0.49° for 50 mm of head movement, half the plan's 1.0° registration budget. The plan's head tracker cancels this by redrawing the symbol for the new eye position, but the correction is only as good as the image distance the software assumes. If the lens is 2% long and nobody measured it, 50 mm of head movement leaves 0.13° of error; a 40 K warm-up leaves 0.08°. Both fit inside the budget, but they are avoidable: measure the image distance once (section 11) and give it to the renderer.

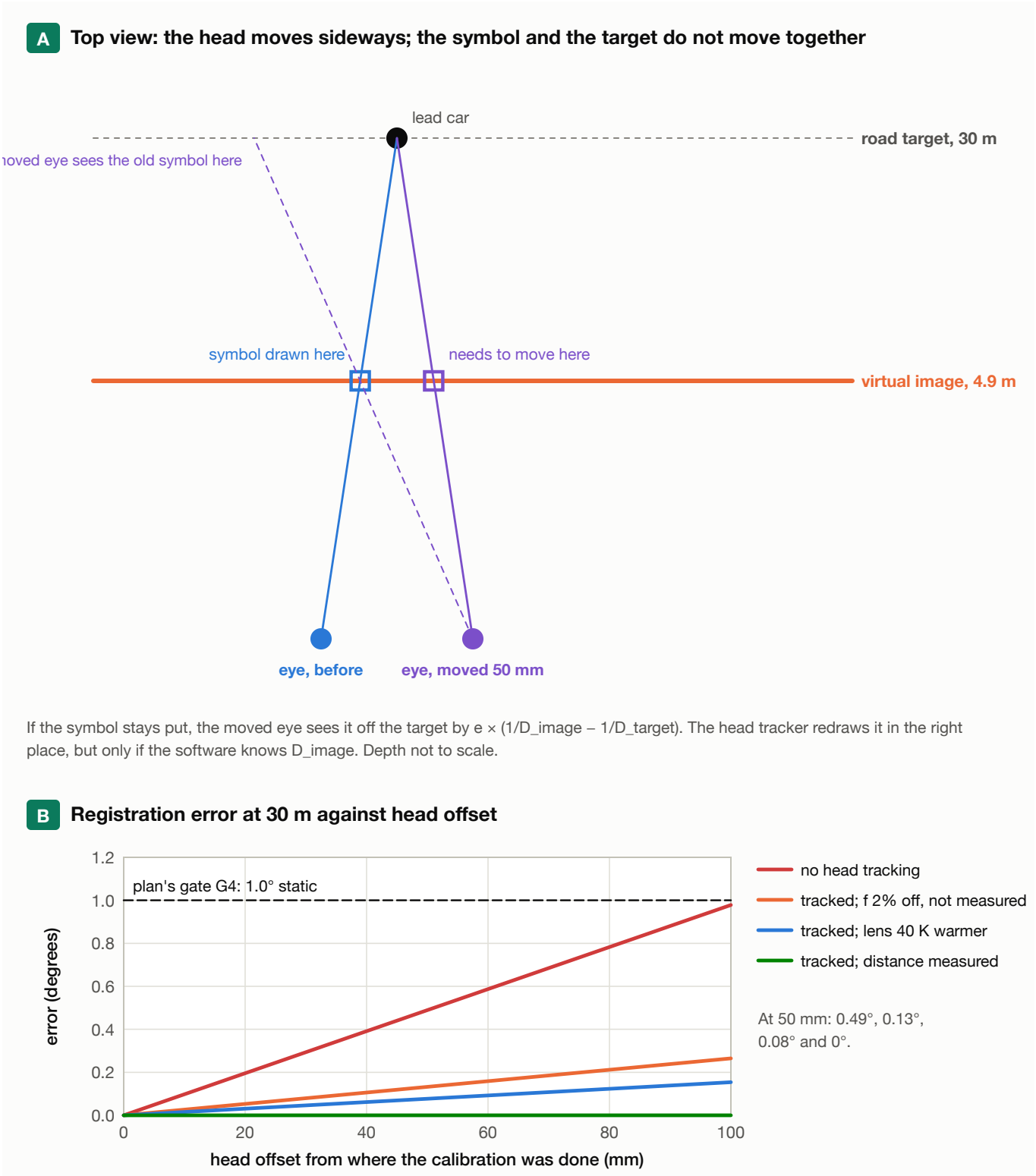
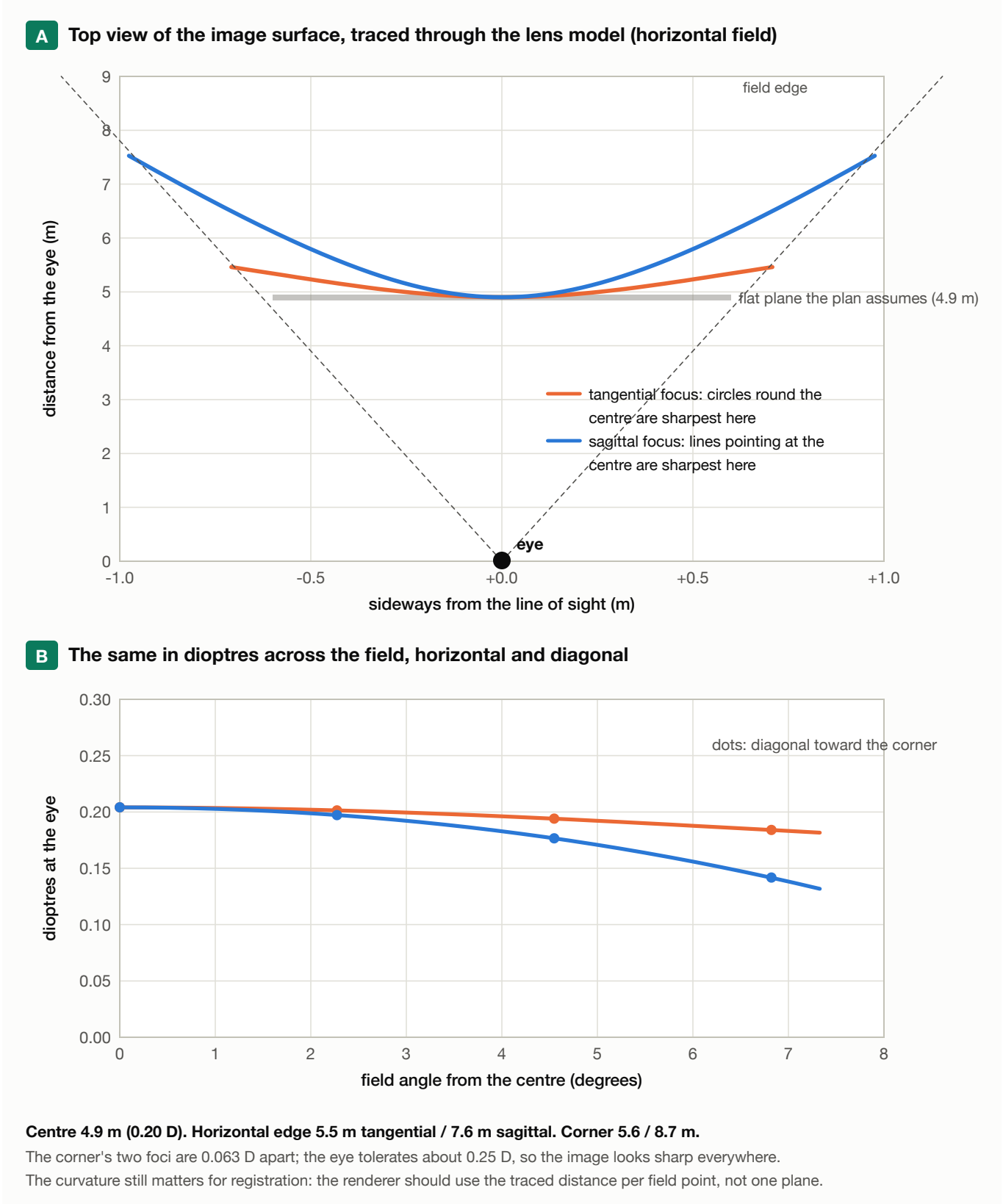


Figure 11. A: why a symbol on the image and a target on the road separate when the head moves (top view, not to scale in depth). B: the registration error at 30 m against head offset, without tracking and with tracking under three assumptions about the image distance.

The image is curved

So far the image has been treated as a flat picture 4.9 m away. The ray trace says otherwise. Toward the edges of the field the light passes the lens at a slant, the effective panel distance grows, and since the panel is close to the focal point the image runs farther away: 4.9 m at the centre, 5.5 m (tangential) to 7.6 m (sagittal) at the side edges and up to 8.7 m at the corners (Figure 12). The two values at each point are astigmatism: lines running around the centre come into focus at one distance, lines pointing at the centre at another.

In dioptres all of this lies between 0.11 and 0.20 D, and the two foci at the corner are 0.063 D apart, well inside the eye's ±0.25 D, so the image looks sharp everywhere. But the parallax argument above applies point by point: a symbol near the edge of the field sits on an image up to 3.8 m farther away than one in the middle. The renderer should use a traced or measured image distance for each part of the field rather than one plane, the same conclusion the project's Cycles checks reached for Stage 2's mirror.



Centre 4.9 m (0.20 D). Horizontal edge 5.5 m tangential / 7.6 m sagittal. Corner 5.6 / 8.7 m.

The corner's two foci are 0.063 D apart; the eye tolerates about 0.25 D, so the image looks sharp everywhere.

The curvature still matters for registration: the renderer should use the traced distance per field point, not one plane.

Figure 12. The traced image surface. A: top view of the tangential and sagittal image surfaces across the horizontal field, with the flat plane the plan assumes. B: the same in dioptres, with diagonal points toward the corner.

5 What the eye sees through the grooves

The HUD's light does pass through the ridges, so it is fair to ask whether the driver sees them. This section answers that in four steps: how far the head can move before the image is cut off, how much of the lens each image point uses, why the ridges blur out, and what remains. The renders at the end show all of it at once.

The eye box

Every head position sees the same virtual image, from a slightly different angle. The light for the left and right edges of the image always comes from the edges of the crop, but it crosses the lens at different places as the head moves. Once one of those rays would have to pass outside the 300 mm lens, that edge of the image disappears (Figure 13). Traced, the whole image stays visible over a box 141 mm wide and 251 mm tall at the eye, larger than the plan's 130 × 70 mm for Stage 2. The combiner's size and the housing opening can cut it further, so check the box once the rig is built.

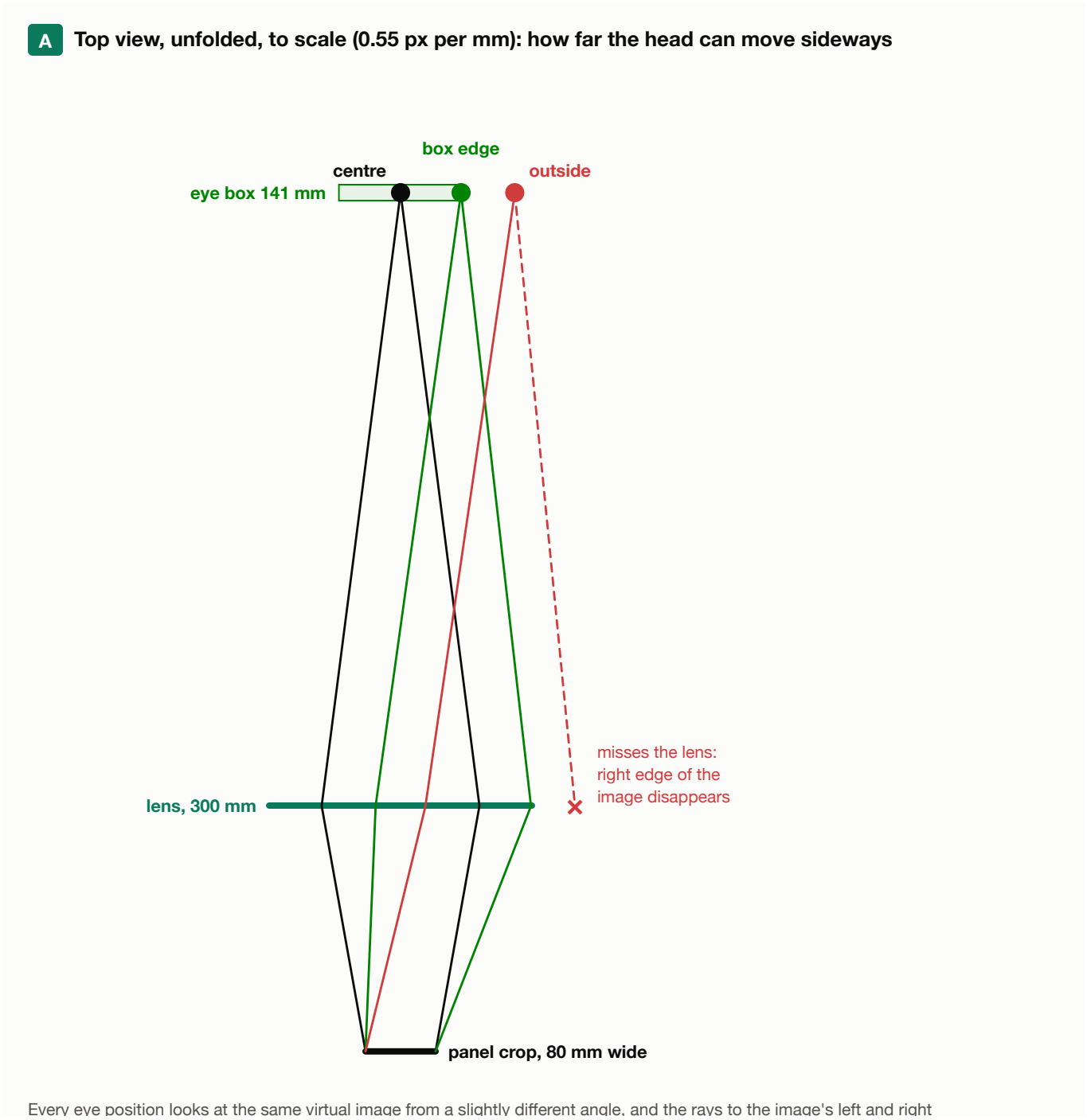
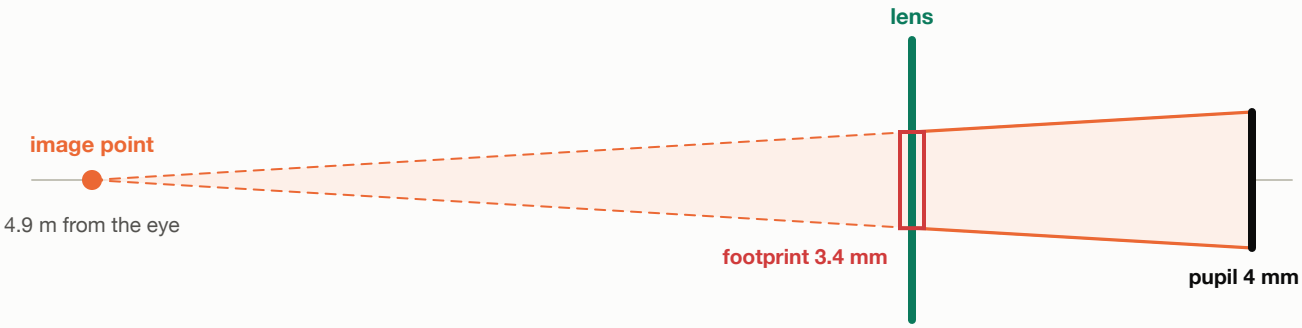


Figure 13. Top view of the unfolded path, to scale, with the rays to the image's left and right edges traced from three eye positions. Outside the eye box one edge ray misses the lens.

Each image point uses a small patch of many grooves

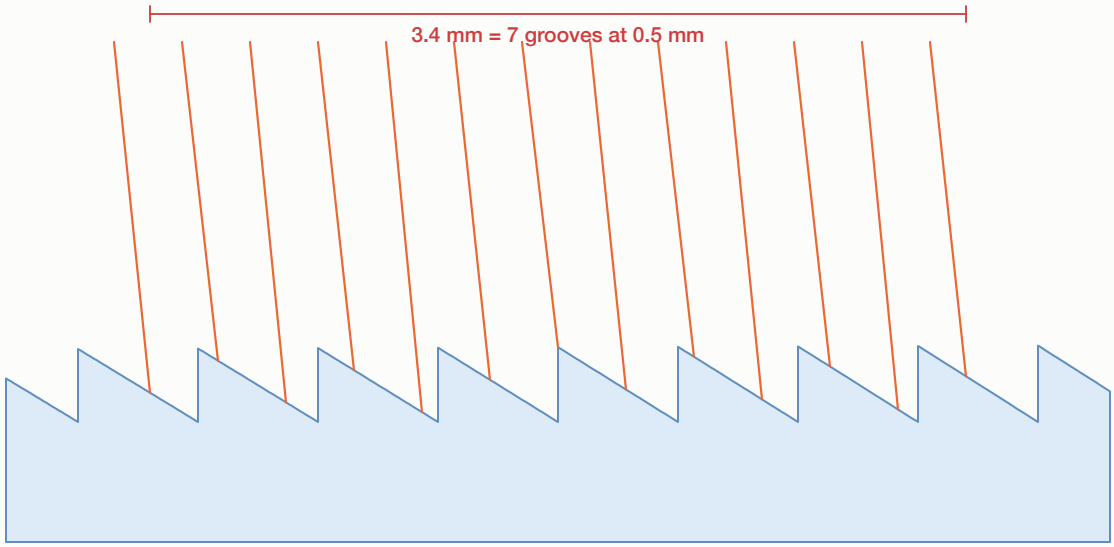
The eye's pupil is 3 to 6 mm wide. The rays from one image point that enter it cross the lens in a patch only a little smaller: 3.4 mm for a 4 mm pupil and 5.1 mm for a 6 mm night pupil (Figure 14). At 0.5 mm per groove, that patch spans 7 to 10 grooves. Each groove delivers its own sliver of the same image point and the eye adds them together, so no single groove, and no single draft wall, decides what a pixel looks like.

A **The light the eye uses for one image point**



The rays that reach a 4 mm pupil from one image point cross the lens in a patch 3.4 mm wide (5.1 mm for a 6 mm night pupil): pupil × image distance ÷ (image distance + eye distance) = 4 × 4.2 ÷ 4.9. Lengths not to scale.

B **That patch on the real teeth near r = 90 mm, magnified 120×: 7 grooves share the work**



Each groove sends its own sliver of the same image point; the eye adds them up. No single groove decides the pixel.

Figure 14. A: the cone of light from one image point to the pupil, and the patch it uses on the lens. B: that patch on the real teeth near the field edge.

The eye also focuses on the image, 4.9 m away, not on the lens 700 mm away. The lens is 1.22 D out of focus, which smears each point on it over about 17′, while the grooves are only 2.5′ apart. A regular pattern seven times finer than the blur nearly vanishes: the traced draft-wall gaps, which a pinhole camera would see as dark lines covering 5% + 10% of the lens at the field edge, become an even dimming with a ripple below 1.5% (Figure 15).



Figure 15. Brightness of a white image across five grooves near the field edge, as a pinhole camera at the eye point would record it and as the eye, focused on the image, sees it. The dark gaps are where sight lines hit draft walls (section 8).

What limits the sharpness

Two effects blur the image across the grooves, and neither depends on the eye's focus (Figure 16). The first is diffraction: each groove is a 0.5 mm wide slit, and light through a slit spreads by about its wavelength divided by the width, 3.2′ here. The second depends on how the lens was made. If each facet is cut as a straight cone rather than following the curved surface it replaces, the image jumps by about the pitch divided by the focal length at every groove: 5.1′ traced for this lens, about 7 pixels of the 5.5-inch panel. A lens with curved facets has almost none (0.5′). Both act across the grooves only, so lines pointing at the lens centre stay sharper than lines running around it.

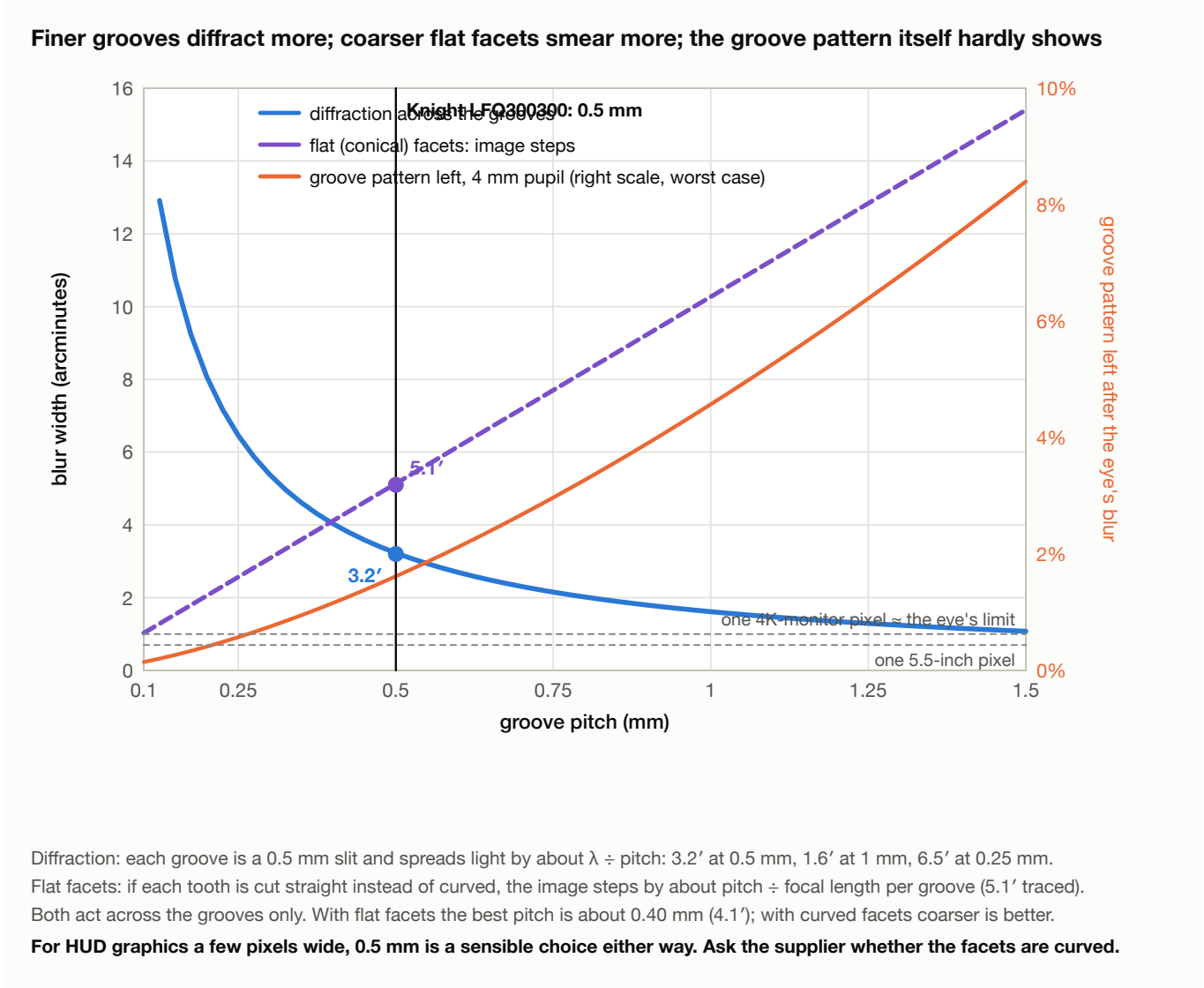


Figure 16. Blur against groove pitch: diffraction, the step from flat facets, and how much of the groove pattern survives the eye's blur (worst case for a 4 mm pupil).

A few arcminutes is coarser than one pixel (0.7–1′) but fine for HUD graphics, whose lines are several pixels wide, and far inside the plan's 1° registration budget. It does mean small text should be drawn larger than on a phone screen. Whether the plan's lens has curved facets is not stated; the supplier can say.

The renders

Figure 1 (on the cover) and Figure 17 are Blender Cycles renders of exactly this situation. The lens is built as real geometry, all 300 teeth out to a radius of 160 mm, each facet carrying the solved curved-surface slope, with PMMA's refractive index. The panel shows a HUD test frame on the 80 × 40 mm crop 280 mm below. The camera sits at the eye point. Each picture is three renders merged, at PMMA's index for red, green and blue light, so colour effects are real. Cycles traces geometric rays, so diffraction is not included.

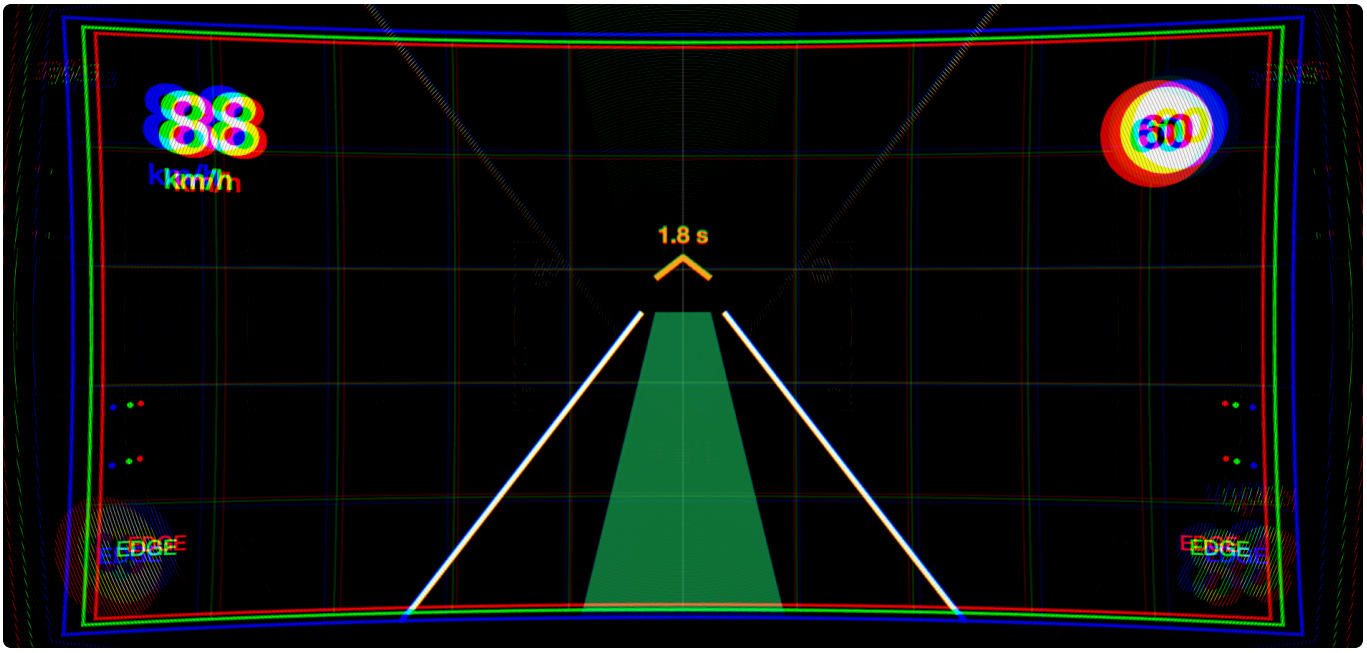


Figure 17. The same view with a pinhole camera instead of an eye: everything in focus, including the lens itself. Compare with Figure 1, rendered with a 4 mm pupil focused at the virtual image.

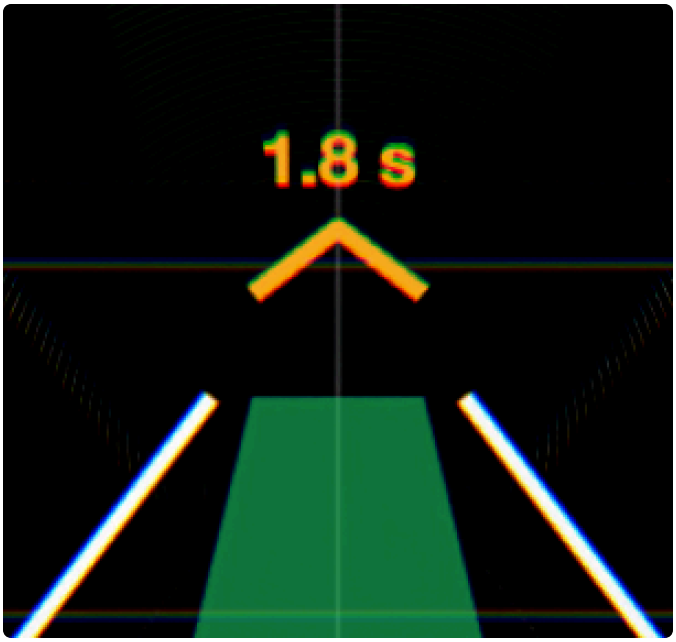


Figure 18. Centre, pinhole, 3× enlarged.

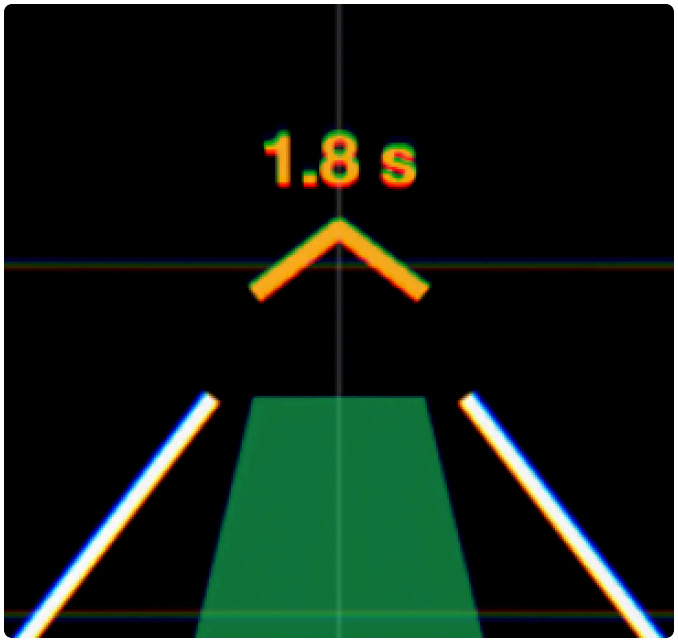


Figure 19. Centre, 4 mm pupil, 3× enlarged.

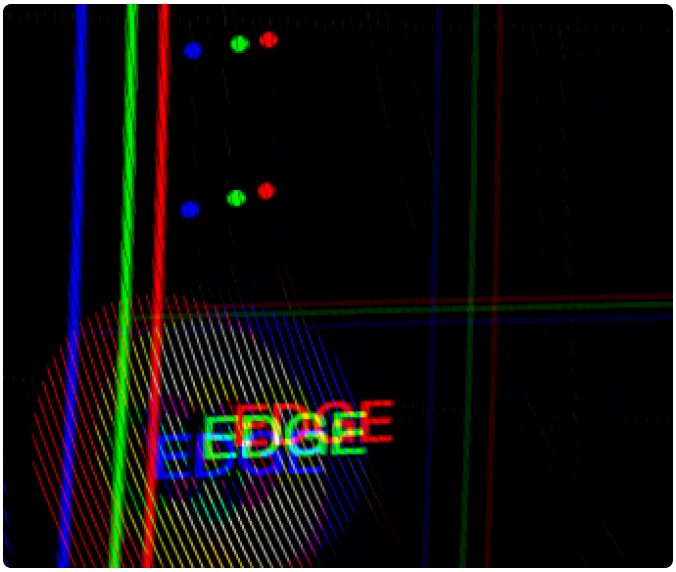


Figure 20. Left edge, pinhole, 3× enlarged.

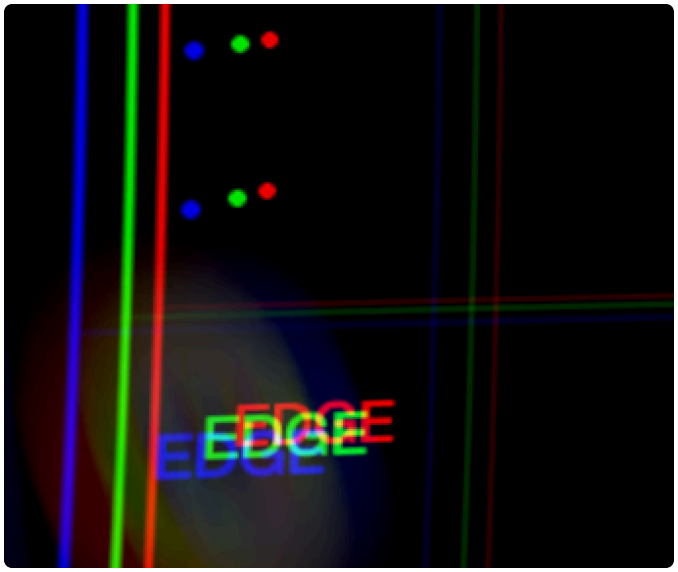


Figure 21. Left edge, 4 mm pupil, 3× enlarged.

What the renders show, from most to least visible:

- **Colour fringes toward the edges.** Blue lands outside red by up to 0.36° at the field edge. This is lateral colour, the subject of section 6, and it is the largest optical flaw in the picture.
- **Bowed straight lines.** The crop border and the grid curve outward at the corners: pincushion distortion, also section 6. The renderer's pre-distortion mesh removes it.
- **Faint copies of bright graphics** in the corners and as dim extensions of the lane lines: reflection ghosts from the lens's own surfaces, about 0.3% as bright as the graphics they copy (section 8).
- **No visible grooves** in the eye render. In the pinhole render, which focuses on the lens as well, the edge shows the fine concentric structure the eye blurs away.

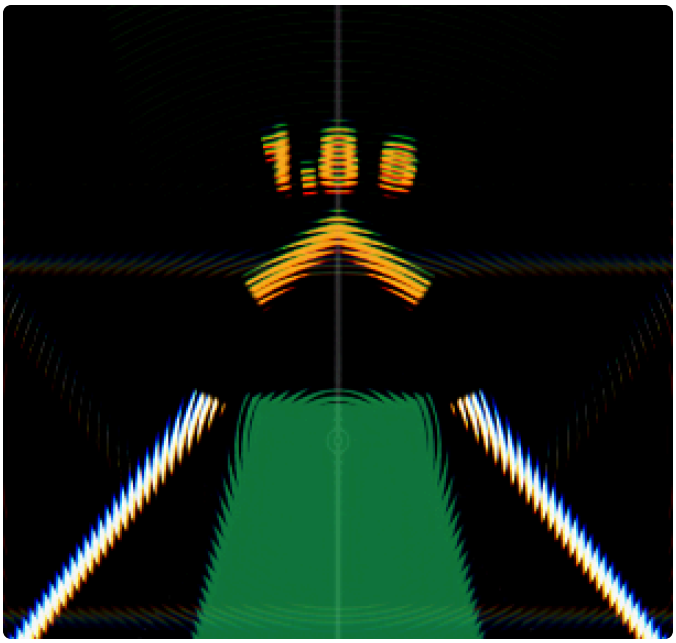


Figure 22. The same centre through a lens with flat (conical) facets, pinhole: the image steps at every groove.



Figure 23. Flat facets seen with a 4 mm pupil: no stripes, but every edge is softened by about 5.1′.

6 Distortion and colour

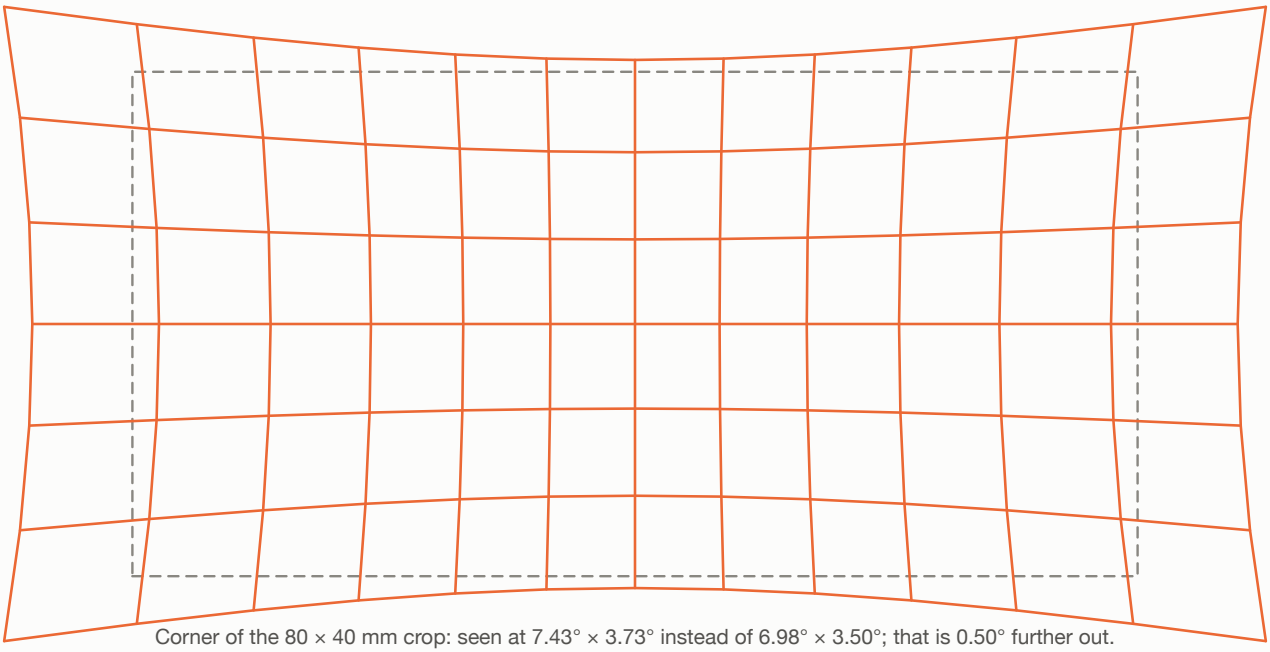
A single lens used as a strong magnifier has two flaws that grow toward the edges of the field. Both are large enough to see, and both are corrected in software rather than optics, which is how the plan already intends to handle distortion.

Pincushion distortion

A magnifier enlarges the edges of the field slightly more than the centre, so a straight grid on the panel looks bowed outward, pulled toward the corners (Figure 24A). Traced, the corner of the 80 × 40 mm crop appears 0.50° further out than a perfect lens would put it. The cure is to draw the content bowed the other way, pulled inward into a barrel shape, by the amounts in Figure 24B: 2.7 mm at the crop’s corner, 5.8% of its distance from the centre. This is the plan's pre-distortion mesh, and the plan measures it directly by photographing a rendered grid from the eye point, so these numbers are a check rather than an input.

A **If the panel shows a straight grid, the eye sees it bowed outward (pincushion)**

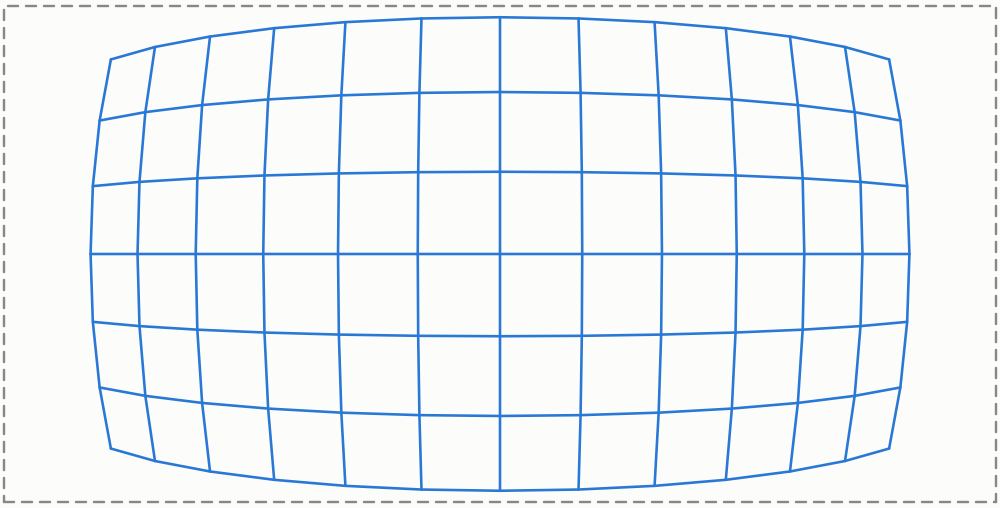
Eye's view, degrees. Dashed: a perfect rectangle. Bending exaggerated 4×; true values below.



Corner of the 80 × 40 mm crop: seen at 7.43° × 3.73° instead of 6.98° × 3.50°; that is 0.50° further out.

B **So the renderer draws the grid pulled inward (barrel) and the eye sees it straight**

Panel, mm. Dashed: the crop. Each grid point is where the traced ray from that direction lands. Exaggerated 4×.



Corner: drawn 2.7 mm closer to the centre (5.8% of its radius); mid-edge 4.8%.

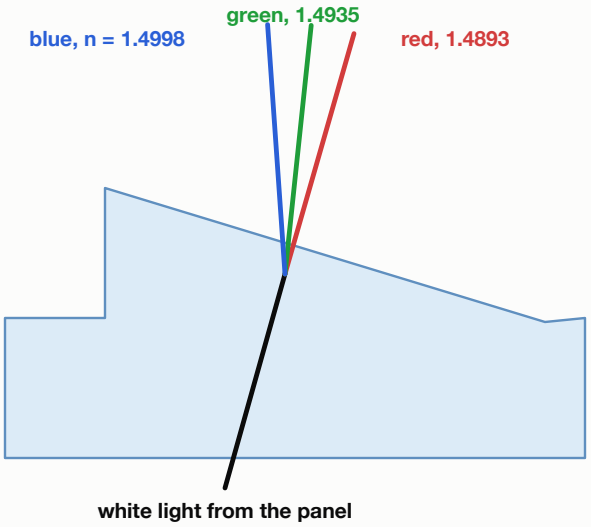
This is the plan's pre-distortion mesh. The calibration photograph measures it directly, so the numbers here are a check, not an input. The mesh is exact for one eye position; within the eye box the distortion changes a little, which the head tracker can follow with a set of meshes measured at several eye points.

Figure 24. A: the grid the eye sees when the panel shows a straight one (pincushion). B: the barrel-shaped grid the renderer draws so the eye sees straight lines. Both traced, with the bending exaggerated four times.

Lateral colour

PMMA bends blue light slightly more than red (its refractive index is 1.4998 at 450 nm and 1.4893 at 620 nm). At every place where the lens bends light, blue is bent a little further, and the lens bends hardest toward its edges. So the blue copy of the picture comes out a little larger than the red one, and a thin white line near the field edge spreads into a small rainbow, blue on the outside (Figure 25). Traced, the split is 0.16° halfway out and 0.36° at the field edge: 21′, or 31 pixels on the 5.5-inch panel.

A Why colours split: the plastic bends blue light more than red

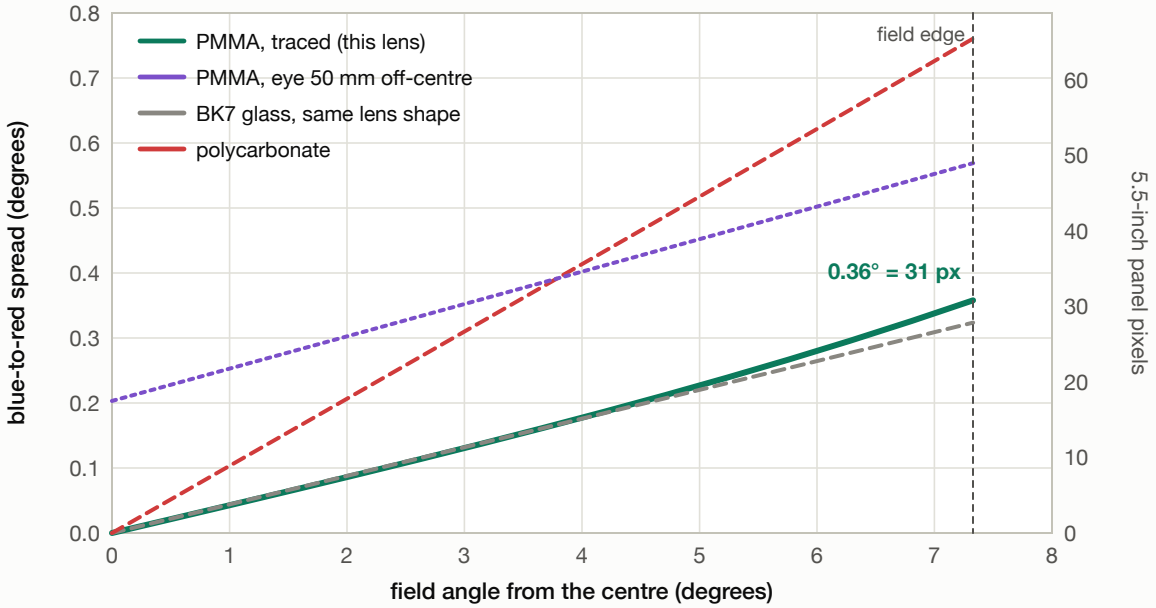


The fan is drawn 25× wider than it is. PMMA's refractive index is 1.4998 for blue and 1.4893 for red, so blue is bent about 2% more at every surface.

The split grows with how hard the lens bends the light at that spot, which grows with distance from its centre.

Abbe number (higher = less colour):
PMMA 58, BK7 glass 64, polycarbonate 28

B Blue-to-red spread across the field, seen from the eye



C What it looks like, and the fix

A thin white line near the field edge spreads into a 21′ rainbow: red on the inside, blue on the outside.

Green-only graphics barely show it (one primary); white text and mixed colours do.

Fix: draw red, green and blue through slightly different pre-distortion meshes (VR headsets do this for their Fresnel lenses). The split depends on where the light crosses the lens, so it moves with the head: with the eye 50 mm off-centre the centre of the field already splits by 0.2°. The per-colour mesh should follow the head tracker, like the geometric one.

Figure 25. A: why colours split at a refracting surface (fan exaggerated). B: the traced blue-to-red split across the field for this PMMA lens, with the same lens shape in BK7 glass and polycarbonate for comparison, and with the eye 50 mm off-centre. C: what it looks like and the fix.

Glass would not fix it: BK7 gives 0.32° against PMMA's 0.36°. Polycarbonate would double it (0.76°), which is one reason PMMA is the right plastic. The fix is in software, as in VR headsets: draw the red, green and blue images through three slightly different pre-distortion meshes, so that after the lens they land on top of each other. Two cautions. The split depends on where the light crosses the lens, which moves with the head: with the eye 50 mm off-centre, even the middle of the field splits by about 0.2°, so the colour meshes, like the geometric one, should follow the head tracker. And an LCD's primaries are bands about 40–60 nm wide, not single wavelengths, so each colour still smears by up to 39% of the uncorrected split (0.14° at the field edge, for typical colour-filter bands, ASSUMED) after the correction.

7 Which way the grooves face

A Fresnel lens works in either orientation, but not equally well. Edmund Optics' rule is that the grooved side should face the longer of the two distances the lens works between. In Stage 1 the short side is the panel, 280 mm away, and the long side is the eye and the virtual image beyond it. So the grooves face up, toward the combiner, and the flat side faces the panel. This is the orientation the lens was designed for, if its design conjugate is the usual one (parallel light on the grooved side).

Figure 26 shows why, with rays from the centre of the panel traced through the real teeth near the lens edge. With the grooves toward the eye, both surfaces share the bending: the flat face takes a little, each facet the rest, at moderate angles. Turned over, the flat face is now on the eye side and leaves almost all the bending to the facets, which meet the light at grazing angles. Much of it reflects, a large share strikes the draft walls and is trapped and guided sideways inside the sheet, and what does come through is badly focused.

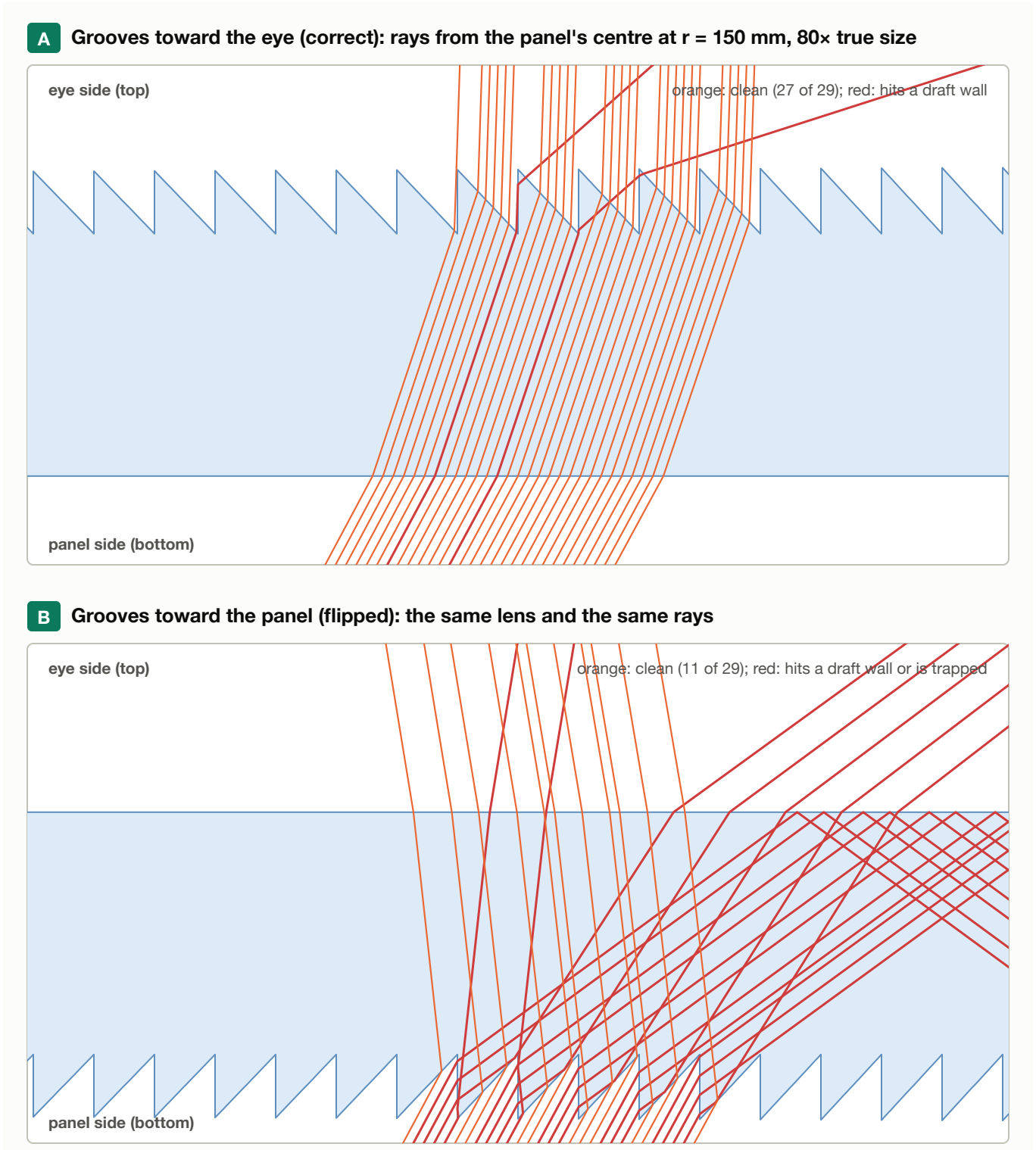


Figure 26. Rays from the centre of the panel through the real teeth at r = 150 mm, at the same scale across and in depth, with the grooves toward the eye (A) and toward the panel (B).

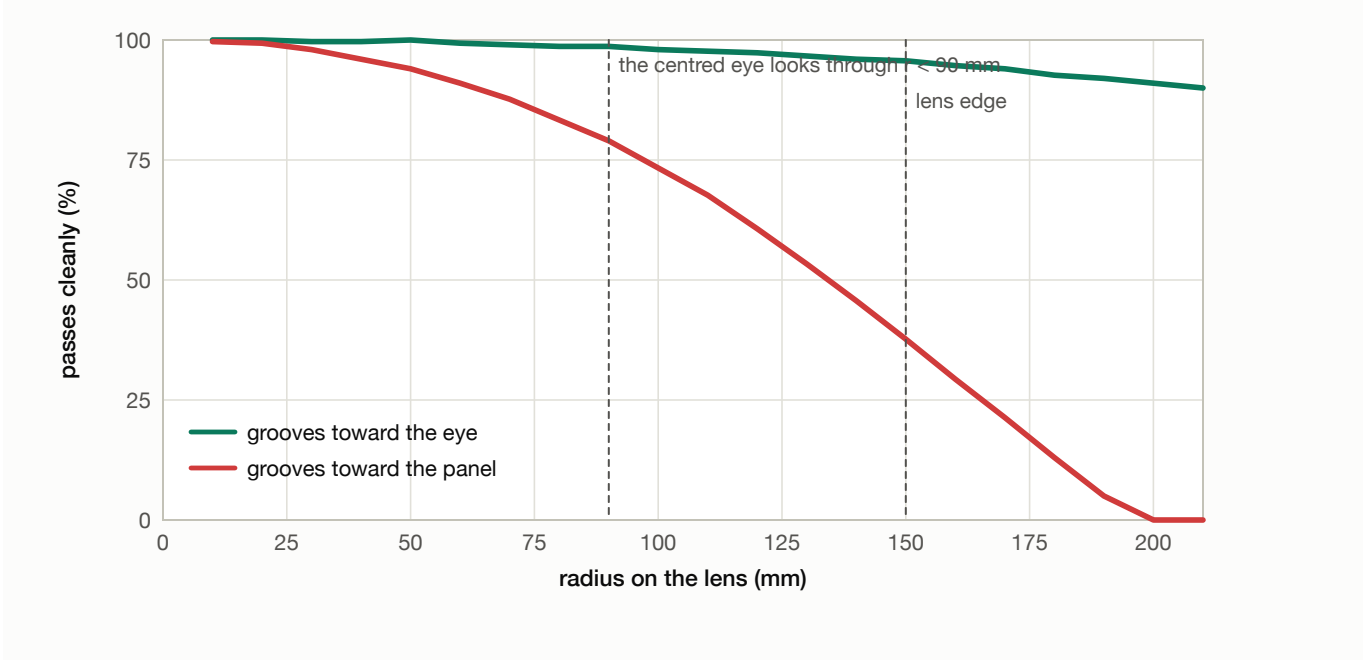


Figure 27. Share of the light from the panel's centre that passes cleanly through the teeth, against radius, for both orientations (groove-level trace).

	GROOVES TOWARD THE EYE	GROOVES TOWARD THE PANEL
Passes cleanly at r = 100 / 150 / 200 mm	98% / 96% / 91%	73% / 38% / 0%
Reflection loss at r = 150 mm (two surfaces)	91.3% transmitted	72.2% transmitted
Steepest angle of incidence at r = 150 mm	28°	75°
Image distance at the field edge, tangential / sagittal	5.5 / 7.6 m	1.1 / 17.8 m
Astigmatism at the field edge	0.05 D	0.85 D
Total internal reflection begins at	not reached	r ≈ 198 mm

Table 2. The two orientations compared, from the ray traces in Appendix A.

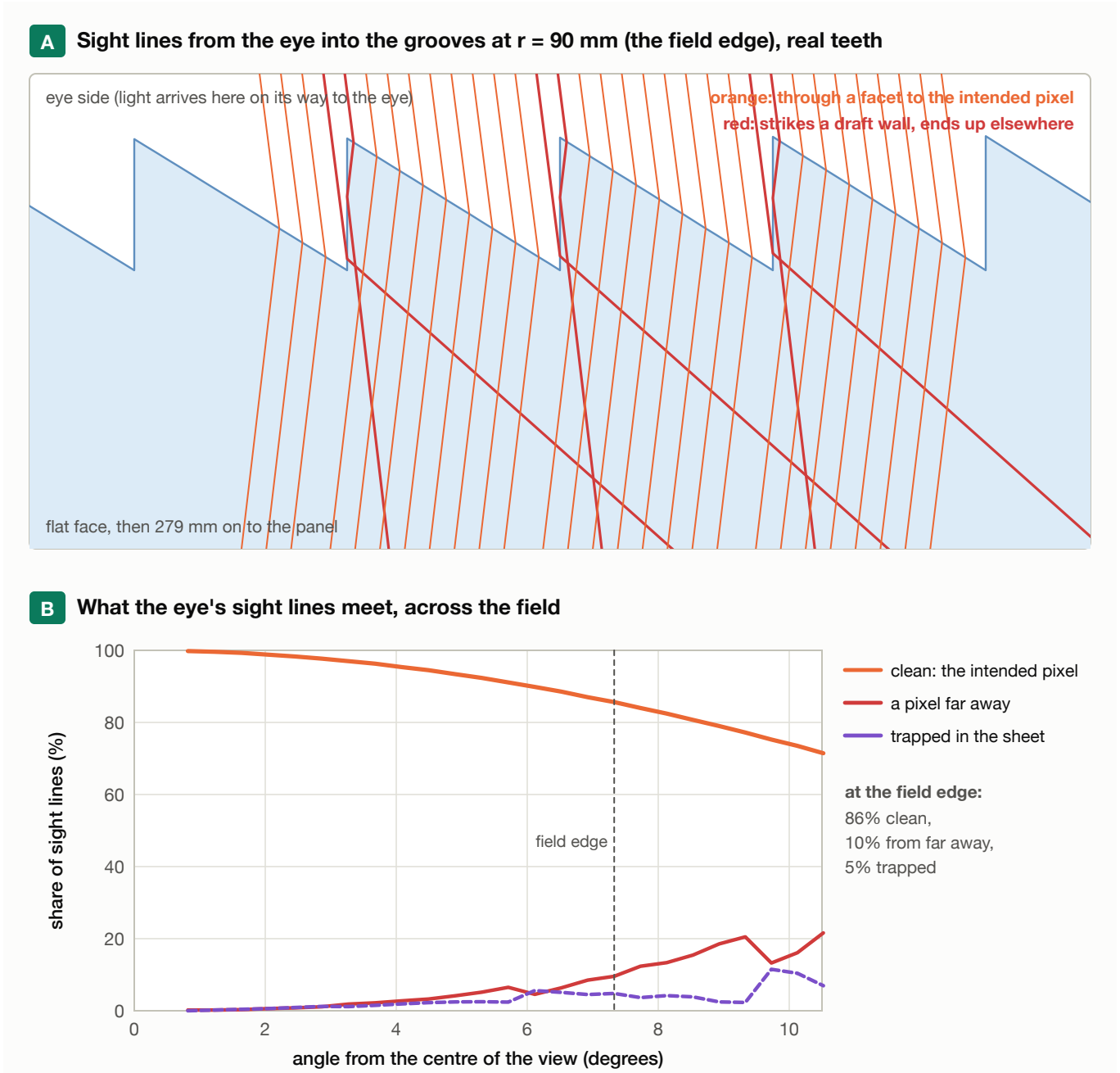
The astigmatism is the decisive number. With the grooves toward the eye the two foci at the field edge are 0.05 D apart, invisible. Flipped, they are 0.85 D apart, which the eye sees as a smear along one direction at the edges of the picture. The lens also looks different in the hand: the grooves feel rough, the flat side smooth. Mount it rough side up, and since the upward-facing grooves will collect dust, keep the box's lid closed when the HUD is off (section 10 gives a stronger reason).

8 Rings, ghosts and glare

Section 5 showed that the grooves themselves blur away. What the eye can still notice comes from light taking the wrong path, and almost all of it starts at a draft wall or at a surface reflection. At night, with bright symbols on a black background, these are the artefacts a passenger will point at.

Sight lines that meet a draft wall

Figure 28 traces sight lines backward from the eye into the real teeth at the edge of the field. Most enter a facet and continue to the intended pixel. Those that arrive near the foot of a tooth meet the next draft wall instead, or are turned by one inside the tooth. They refract at a steep angle, are trapped by total internal reflection at the flat face, and either come out far across the lens or never leave the sheet. At the field edge 86% of the sight lines are clean, 10% end on a panel spot far from the intended pixel, and 5% are trapped; at 2.5° from the centre 98% are clean.



Dimming toward the edge. The wall zones repeat every 0.5 mm, so the eye's blur turns them into an even loss: about 14% darker at the field edge than at the centre. It does not depend on the pitch: finer grooves are shallower by the same factor.

Ghosts and rays. The 'far away' light comes from a panel spot well outside the crop. If that spot is lit (a bright graphic, the stock backlight, sunlight on the panel), the eye sees a faint smear arranged around the lens centre: rings and radial streaks.

At night this is the visible part: a dim glow or streak next to bright symbols on black. Keep the panel black outside the graphics, light only the crop, and make every surface the stray light can reach matte black. Dust in the grooves scatters the same way.

Figure 28. A: sight lines from the eye into the real teeth at r = 90 mm, the field edge. B: what the eye's sight lines meet across the field, from a trace of 2,000 sight lines every 5 mm of radius.

Three consequences follow.

- **An even dimming toward the edges.** The wall zones repeat with every groove, so the eye's blur spreads them into a smooth loss: the field edge is about 14% darker than the centre. The fraction does not

depend on the pitch, because finer grooves are also proportionally shallower. A brightness ramp in the renderer can flatten it if it shows.

- **Faint smears from bright spots elsewhere.** The "far" sight lines see a panel spot well away from the one they should. If that spot is lit, its light appears where it should not, arranged around the lens centre, which is why Fresnel artefacts look like arcs, rings and radial streaks. The remedy is to keep everything outside the graphics black: the crop's backlight only behind the crop, the rest of the panel and the box interior matte black.
- **Dust and scratches** on the grooved face scatter in the same way. The grooves face up in Stage 1, so a closed lid when the HUD is off and a soft brush, not a cloth, keep them clean.

Reflection ghosts

Every uncoated surface reflects about 4% of the light that meets it head-on, more at slant. Light from the panel that reflects off the grooved face, back down, and off the flat face again leaves the lens a second time on a different path and forms a faint, displaced copy of the bright graphics. In the renders these ghosts appear in the corners and as dim continuations of the lane lines; raising the exposure sixteen-fold makes them obvious (Figure 29). Measured in the render, a ghost of a thin white line is about 0.3% as bright as the line, and the ghost of a large filled area like the ribbon about 1% of that area's brightness; black parts of the picture glow at about 0.1% of white from scattered light. That is invisible by day and faint at night, but noticeable against black if the HUD is turned up. An anti-reflection coating, which Knight Optical offers on its stock lenses, reduces each surface's 4% to well under 1% and the ghosts by more than an order of magnitude; it is worth having if Stage 1 is shown at night.

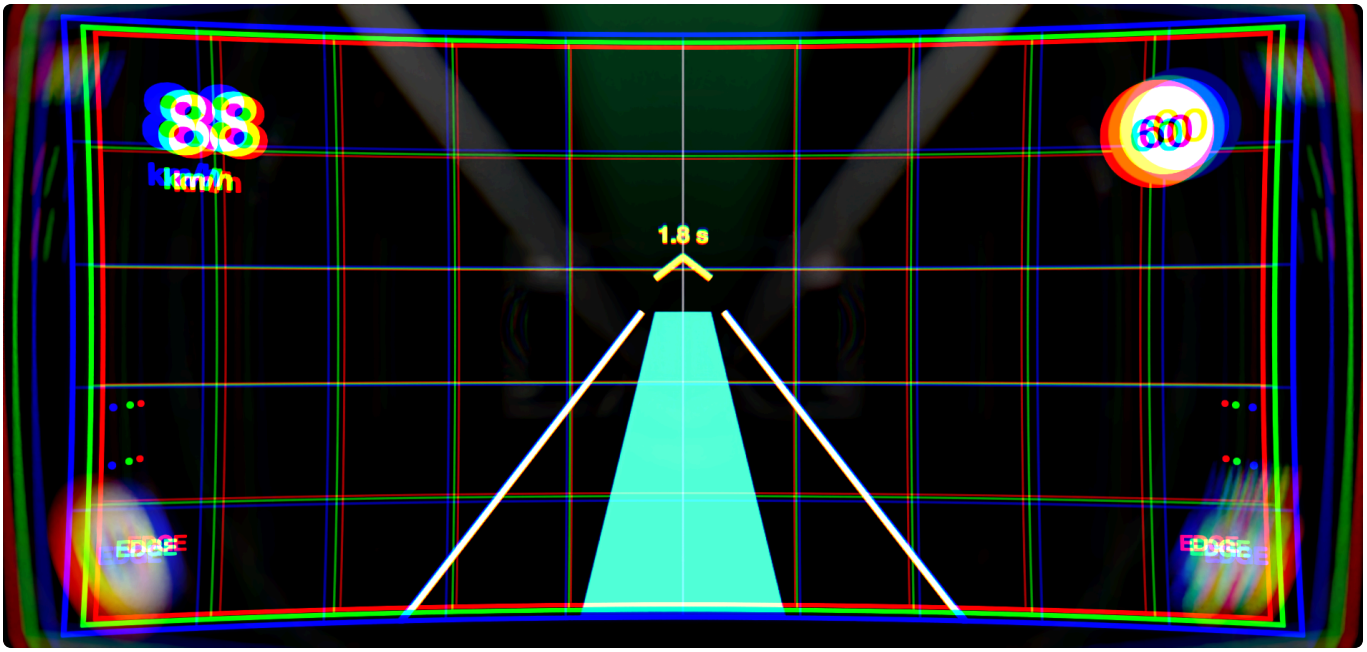


Figure 29. The eye render of Figure 1 with its exposure raised 16×. The extra copies of the speed, the sign and the lane lines are reflection ghosts from the lens's two surfaces; with the reflections switched off in the renderer they disappear.

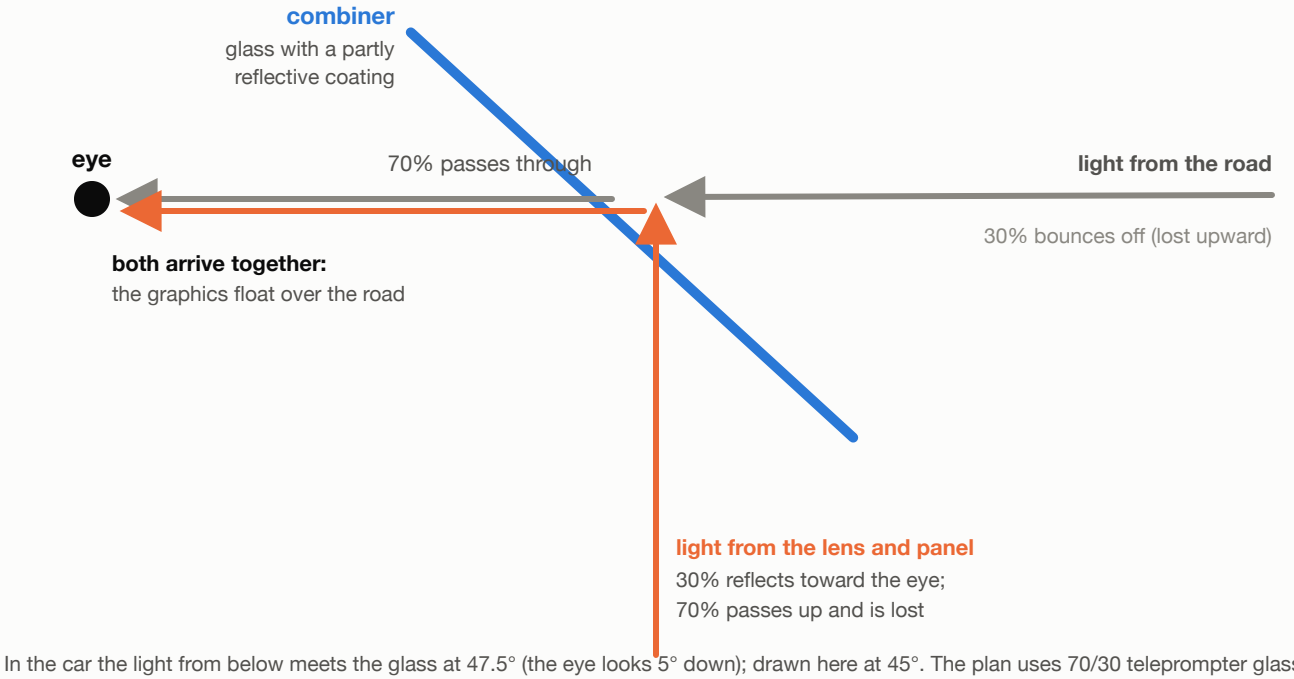
Moiré

Two fine regular patterns laid over each other can produce a coarse beat pattern. Here the candidates are the grooves (2.5′ apart from the eye) and the panel's pixels (0.7–1′ apart). The eye render shows no moiré: the eye's blur over 7 or more grooves removes the groove pattern before it can beat with anything. (The pinhole render, which sees the grooves sharply, does show striped patterns where a ghost overlaps them.) If it ever appears, for example on a camera recording the HUD with a small aperture, changing the panel distance slightly or rotating the lens a few degrees moves or removes it.

9 The combiner, and how bright the image is

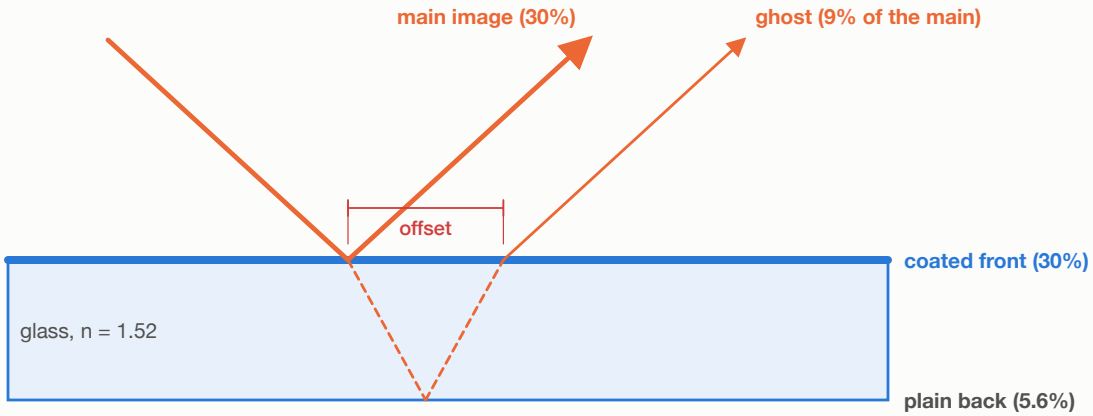
A combiner is a sheet of glass that is partly a window and partly a mirror. The plan uses 70/30 teleprompter glass: a coating on one face reflects 30% of the light that meets it and lets 70% through. Tilted above the lens, it does two jobs at once (Figure 30A). Light from the road passes through it to the eye, dimmed to 70%. Light from the lens below reflects off it toward the eye, 30% of it. Both arrive along the same line of sight, so the graphics appear to hang over the road: the two pictures are *combined*. The rest of the HUD light (70%) carries on upward through the glass and is lost, and 30% of the road light reflects away.

A **A sheet of glass that is partly a window and partly a mirror**



In the car the light from below meets the glass at 47.5° (the eye looks 5° down); drawn here at 45°. The plan uses 70/30 teleprompter glass.

B **The back surface reflects too: a second, fainter copy, slightly shifted**



For 3 mm glass the two copies are 2.2 mm apart, which at 4.9 m is 1.6′; 6 mm glass doubles it (3.2′). A pixel is about 0.7–1′. The ghost carries 9% of the main image's brightness for unpolarised light. At 47.5° the plain surface reflects 10.6% of s-polarised light but only 0.6% of p-polarised, and the LCD's light is polarised: turned to p, the ghost almost vanishes, but the coating's own reflectance changes with polarisation too. Measure both with the panel at the combiner before deciding (section 9).

Figure 30. A: what a combiner does. B: the second, fainter reflection from its uncoated back surface, and how far it is offset.

The ghost from the back surface

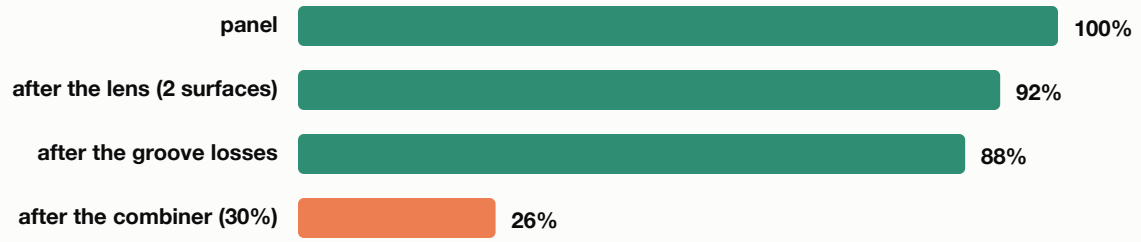
The coating is on one face; the other face is plain glass and reflects too, about 5.6% at this angle for unpolarised light. That second reflection comes from slightly further along, so it makes a second, fainter copy of the HUD, shifted by twice the glass thickness times a geometric factor (Figure 30B). For 3 mm glass the copies are 2.2 mm apart, which at 4.9 m is 1.6′, about two pixels; for 6 mm glass, 3.2′. The ghost carries about 9% of the main image's brightness. This is the same physics as the windshield ghost the plan analyses for Stage 2, at a much smaller scale because the combiner is thin and the image is far.

The LCD's light is polarised, which changes both reflections. At 47.5° the plain back face reflects 10.6% of light polarised one way (s, vibrating parallel to the glass surface) and only 0.6% of the other (p). Turning the panel, or adding a half-wave film, so its light arrives p-polarised would nearly remove the back-surface ghost, but the coating's 30% is also specified for unpolarised light and may drop for p. Measure the main reflection and the ghost with the actual panel at the combiner, in both orientations, and choose. PMMA's low birefringence matters here: a polycarbonate lens would scramble the polarisation unevenly and make the result patchy. Polarised sunglasses can black out the HUD entirely, as the plan notes for Stage 2.

Brightness at the eye

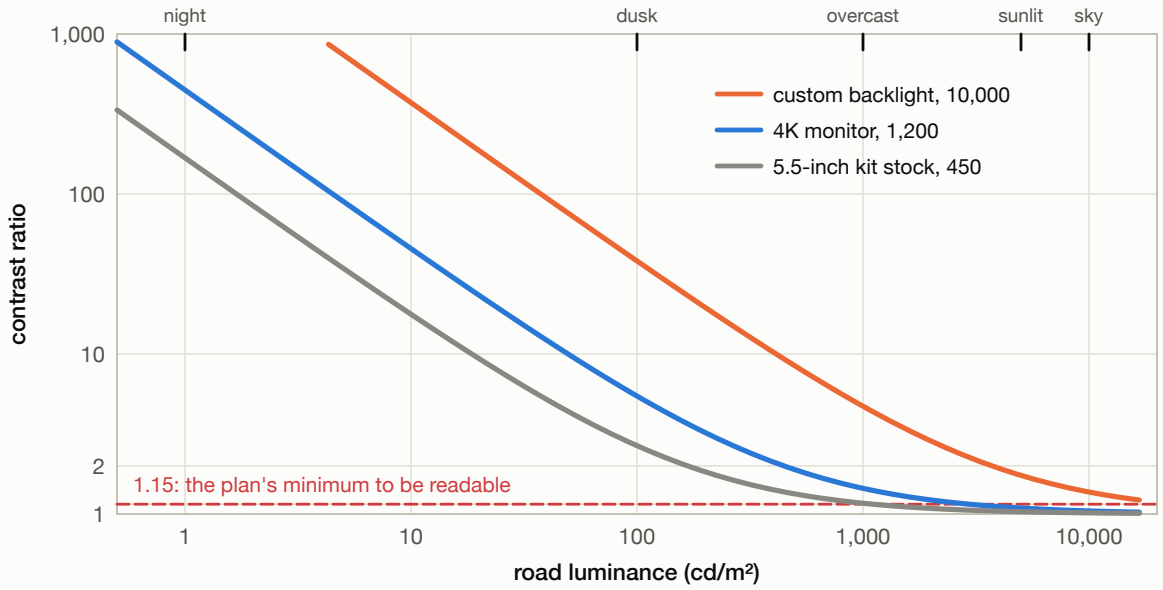
Section 3 explained that the lens does not change luminance except for its losses. So the brightness at the eye is the panel's luminance times three factors (Figure 31A): the lens's two surfaces (92.4% on axis), losses at the grooves and draft walls (5%, ASSUMED near the centre; up to 14% at the field edge, section 8), and the combiner's 30%. Together, 26%. The plan's 4K monitor at 1,200 cd/m² gives 316 cd/m² at the eye; the 5.5-inch kit's stock backlight gives 118; the custom backlight's design 10,000 cd/m² gives 2,633.

A **What reaches the eye: lens, then the combiner's 30% reflection**



A lens does not change luminance (cd/m²) except for its losses, however much it magnifies. So: eye = panel × 26%.
16-inch 4K monitor, 1,200 cd/m²: 316 cd/m² at the eye. 5.5-inch kit, stock: 118. 5.5-inch kit, custom backlight, 10,000: 2,633.

B **Contrast against the road seen through the combiner (70%)**



Night (about 1 cd/m²) is easy with anything: the HUD must be dimmed 53× from full for the plan's 50 cd/m² night ceiling.
With the custom backlight, Stage 1 stays readable even against a sunlit road. The limit in daylight is sunlight through the lens (section 10).

Figure 31. A: the fraction of the panel's luminance that reaches the eye. B: contrast of the HUD against the road seen through the combiner, for three panels, against the road's luminance, with the plan's 1.15 readability threshold.

Against the road (Figure 31B), any of the three is far too bright at night, about 1 cd/m², and must be dimmed: the custom backlight by 53× to reach the plan's 50 cd/m² night ceiling. In overcast daylight the monitor and the custom backlight are readable; the stock kit is marginal. With the custom backlight, Stage 1 would even stay above the threshold against a sunlit road, partly because the combiner dims the road to 70% while reflecting 30% of the HUD's light, where Stage 2's windshield reflects only 24%. Brightness is therefore not what limits Stage 1 to night and overcast use. Sunlight is.

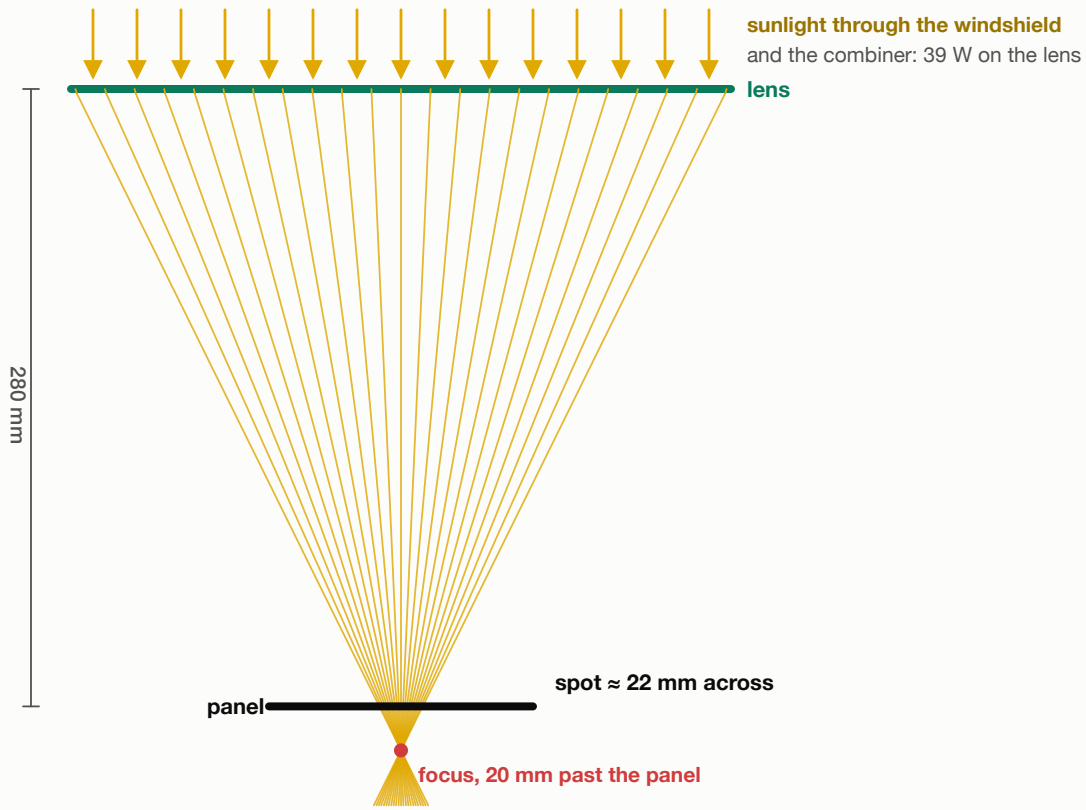
10 Sunlight through the lens

A 300 mm lens with a 300 mm focal length is, among other things, an excellent burning glass. Pointed at the sun, it gathers the light falling on 0.09 m² and brings it to a point. In Stage 1 the panel sits 20 mm short of that point, which is close enough to matter. The plan treats the sun hazard carefully for Stage 2 (cold-mirror fold, infrared filter, polariser, shutter and a sun-watch rule, section 6.8 of the plan). Stage 1 has none of these, and the plan does not discuss its sun exposure.

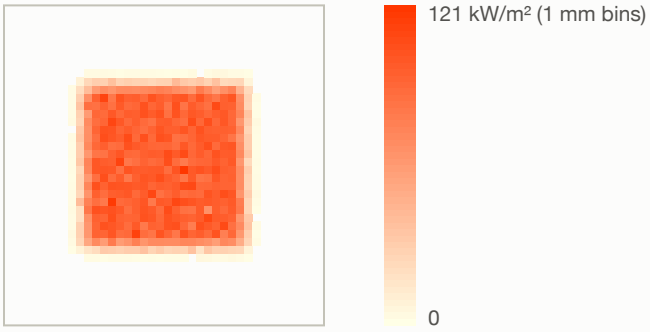
Sun straight down the lens

With the sun on the lens axis, after the windshield's 70% and the combiner's 70% (both ASSUMED), 39 W reaches the lens (55 W if the combiner is not above it). Traced through the real facets, it lands on the panel as a square spot with 90% of the power within 11 mm of its centre, peaking at 121 kW/m², about 121 times full sunlight (Figure 32). For comparison, Texas Instruments designs production HUDs for at most 40 kW/m² on the picture generator, and the plan cites a BMW HUD whose TFT warmed 45 °C at 12 kW/m². An LCD's front polariser absorbs about half of the light that hits it, so roughly 20 W would be dumped into a few square centimetres of the panel.

A **Side view, to scale (1.1 px per mm): sun straight down the lens axis**



B **The spot on the panel, traced: power per area (kW/m²)**



Collected	**39 W** (55 W with no combiner above)
Spot	**90% within 11 mm** of its centre
Peak	**121 kW/m²** about 121 suns
TI limit	**40 kW/m²** production HUD design peak
BMW TFT	**+45 °C at 12 kW/m²** (plan, section 6.8)

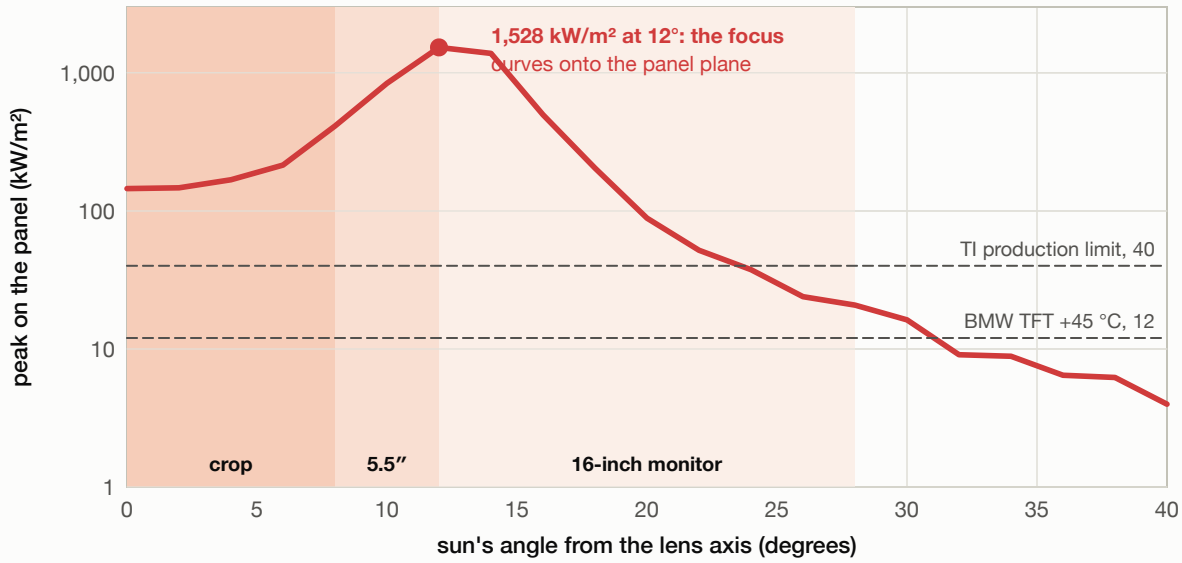
There is no cold mirror, IR filter or shutter in Stage 1: all of this lands on the LCD, whose polarisers absorb half of it at once.
The windshield's 70% and the combiner's 70% transmission are ASSUMED; the lens is traced with its real facets.

Figure 32. A: sunlight straight down the lens axis, traced through the facets to the panel and beyond the focal point (to scale). B: the traced spot on the panel, power per area.

Sun off the axis: worse, not better

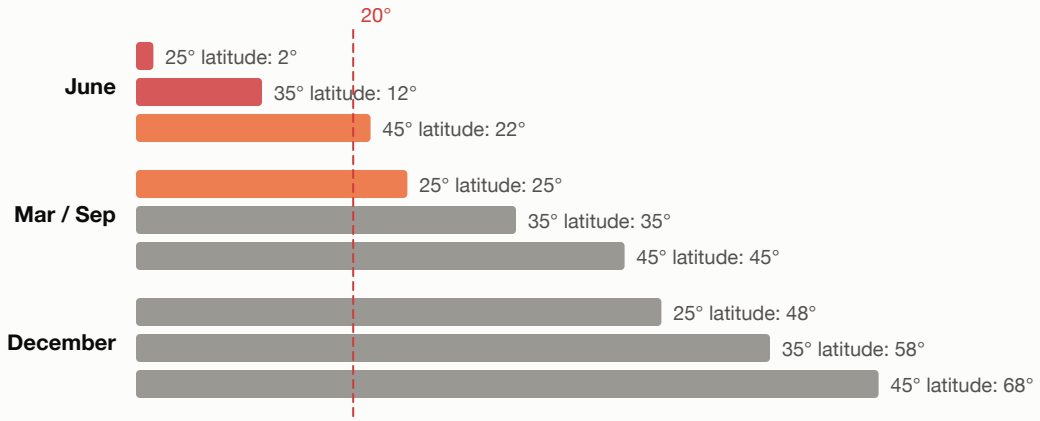
Tilting the sun moves the spot across the panel: about 52 mm at 10° and 112 mm at 20°. One might expect it to spread as it moves. It does at large angles, but first it sharpens (Figure 33A). Off the axis the lens's focal surface curves toward it (the same field curvature as in section 4, seen from the other side), and around 12° it crosses the panel plane. There the sun is nearly in focus on the panel: the traced peak reaches 1,528 kW/m² at 12° (in 1 mm bins), enough to burn through a polariser in seconds and to char most plastics. Up to about 28° the spot still lands on the 16-inch monitor, and up to 12° on the 5.5-inch panel.

A **Sun off the lens axis: the spot moves across the panel and gets sharper before it spreads**



Shaded bands: where the spot's centre lands (on the crop up to 8°, on the 5.5-inch panel up to 12°, on the monitor up to 28°).
Off axis the focal surface bends toward the lens (section 4) and crosses the panel plane near 12°, so the worst case is not straight down.

B **Sun's angle from straight up at solar noon, with the lens lying flat (lens axis vertical)**



Summer middays at most latitudes put the sun within 20° of straight up for hours: a parked car with the rig uncovered will eventually focus it onto the panel. The real angle also depends on the car's heading, the lens tilt and the roof line, so the plan's sun-watch rule (section 6.8 of the plan) has to be extended to Stage 1, with the lid as the primary defence.

Figure 33. A: peak power on the panel against the sun's angle from the lens axis, with where the spot lands. B: the sun's angle from straight up at solar noon for three latitudes and three seasons, with the lens lying flat.

How often does that happen? With the lens lying flat on the dash, its axis points straight up, so what matters is the sun's angle from overhead. At solar noon in June it is 2° at 25° latitude, 12° at 35° and 22° at 45° (Figure 33B). The windshield's rake lets that light in from the front, and the combiner passes 70% of it. A parked car with an uncovered rig will, on a summer day, sweep the spot across the panel. A second path exists too: sunlight from behind and above can reflect off the combiner (30%) straight down the HUD's own light path.

Rules for Stage 1

1. An opaque lid over the lens, closed whenever the HUD is not in use and always when parked. This is the primary defence, and it also keeps dust out of the upward-facing grooves.
2. Never leave the rig in a parked car in the sun, even covered: the dash can pass PMMA's 65–95 °C service limit and warp the lens.
3. Extend the plan's sun-watch rule (sun position from GNSS time, position and heading) to Stage 1 with the patch of Figure 33A, and close the lid and blank the panel when the sun enters it.
4. Put the plan's thermistor on the back of the panel in Stage 1 as well, with the same cut-off.
5. Before any daytime drive, repeat the plan's thermocouple test with the empty box in the sun.

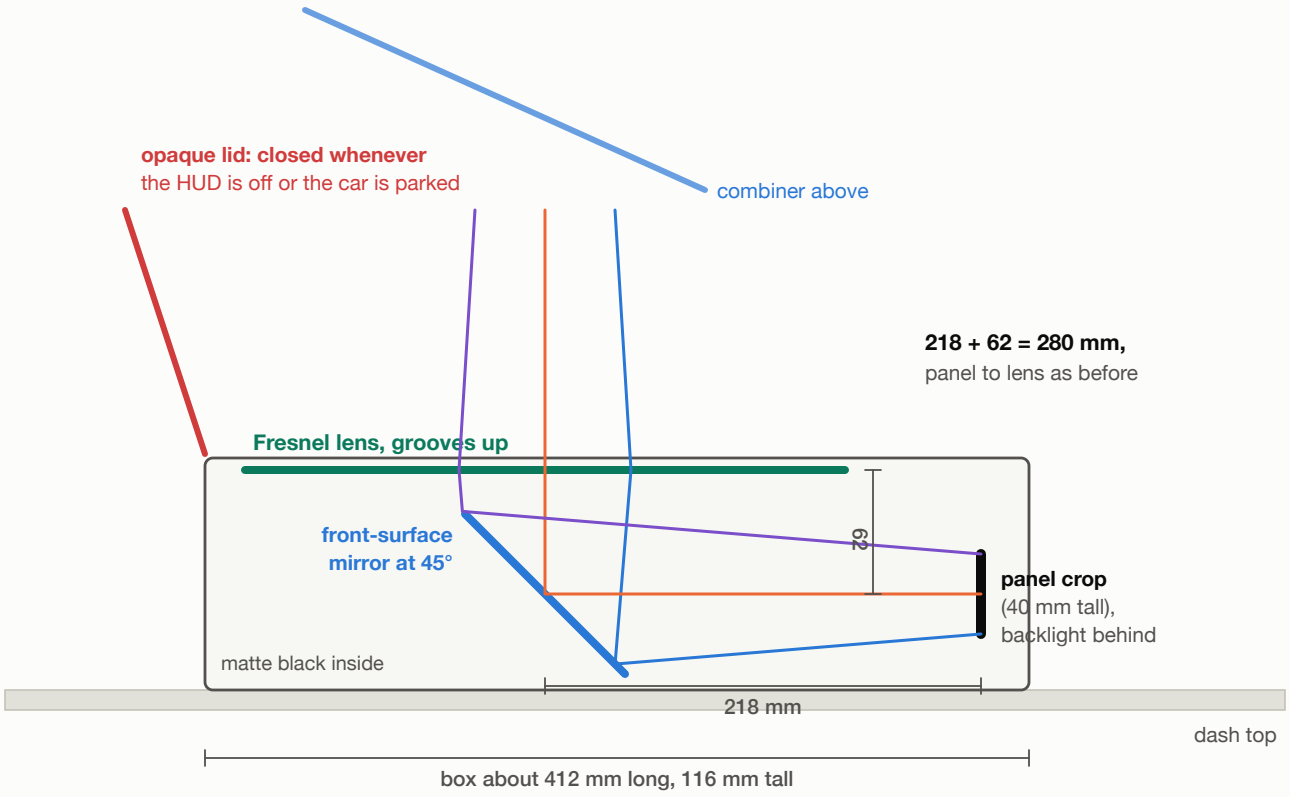
11 Building, aligning and checking it

Packaging: fold the path

With the lens on the dash and the combiner above it, a straight 280 mm drop puts the panel inside the dashboard (Figure 2). A front-surface mirror at 45° under the lens folds the path: the panel stands upright at one end of a low box, its light runs along the box, turns up at the mirror and passes through the lens (Figure 34). The optics are unchanged because the path length is unchanged; the image is mirrored once, which the renderer undoes. The mirror must be front-surface (coated on top) for the same reason the combiner has a ghost: a rear-surface mirror would add a second reflection. This packaging is a suggestion; the plan does not say how Stage 1 is mounted.

A suggestion, not in the plan: fold the 280 mm path so the box sits on the dash

Side view to scale (1 px per mm). Light path: panel, then a front-surface mirror at 45°, then the lens and the combiner. Same path length.



The mirror reverses the image once; the renderer flips it. A front-surface mirror avoids a second, ghost reflection.
Set the 280 mm along the folded path with the panel on a slide (section 11) and keep every stray-light surface matte black.
The mirror and panel must stay put to well under a millimetre: 1 mm of panel movement shifts the image 0.2 m (section 4).

Figure 34. A suggested Stage 1 box, to scale: panel, 45° front-surface mirror, lens on top with the grooves up, and the opaque lid that section 10 requires. Not in the plan.

Setting the panel distance by parallax

Section 4 showed that the focal length is uncertain by ±2% and that the image distance follows it closely, so set the panel distance on the lens actually used. The easiest reference is a physical target at the intended image distance: a pole or a cross on a stand, 4.9 m from the eye, in line with the HUD. Show a cross on the panel at the target's position, look past it at the target, and move the head from side to side (Figure 35). If the cross slides against the head movement, the image is nearer than the target: move the panel away from the lens. If it slides with the head, the image is farther: move the panel toward the lens. When the cross stays on the target, the distance is right to within a few centimetres.

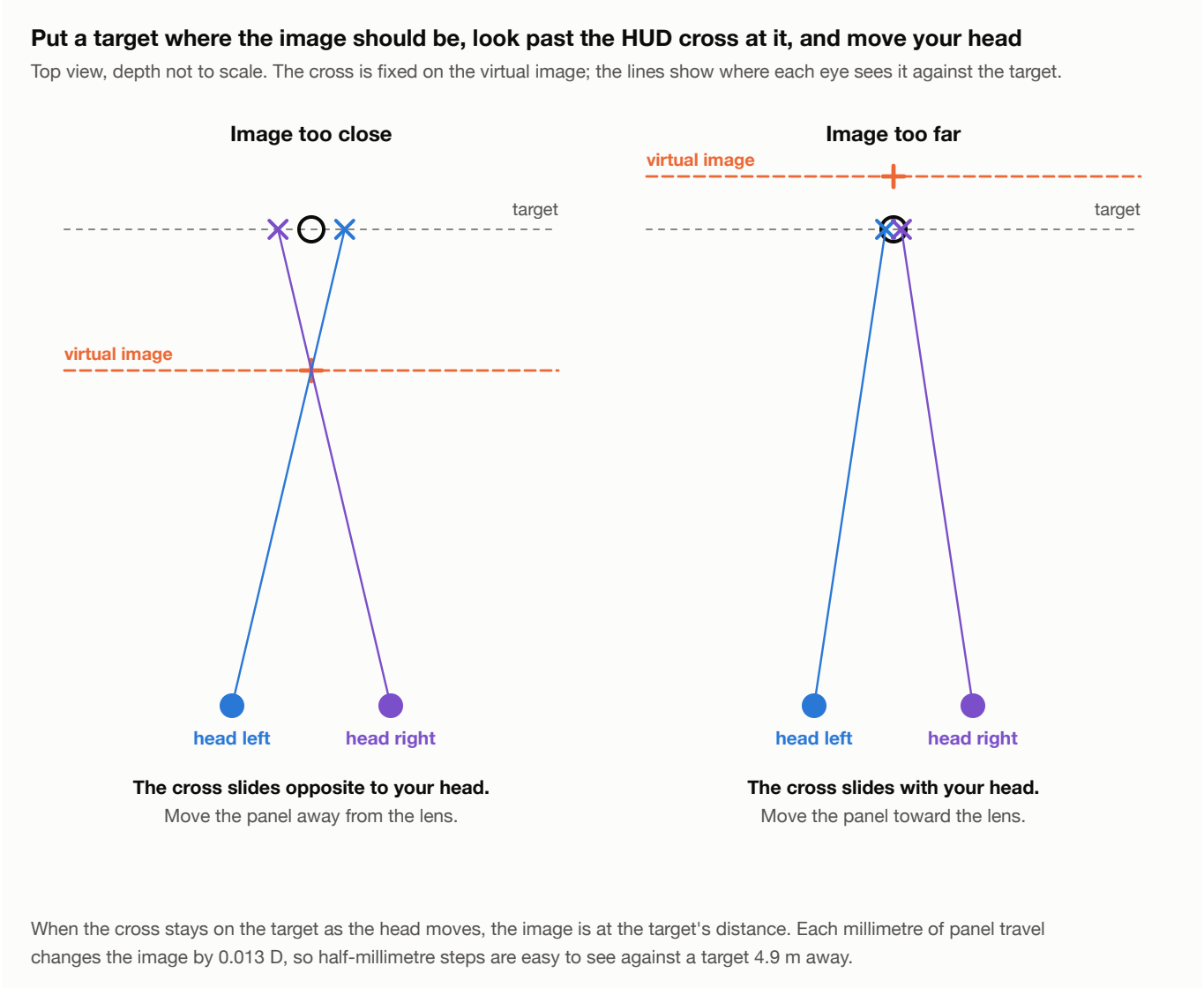


Figure 35. Parallax alignment: where the HUD cross appears against the target as the head moves, for an image too close and too far.

A useful cross-check is the focal length itself. Hold the lens, grooves toward a distant lamp (more than 20 m away; never the sun), and move a white card behind it until the lamp's image is sharpest. The card is then about f – 1.34 mm from the flat face. That number gives the starting panel distance from Figure 10A before the parallax check.

Checklist

1. Lens: precision grade, PMMA, grooves facing up toward the combiner (section 7). Ask the supplier whether the facets are curved and whether an anti-reflection coating is available (sections 5 and 8).
2. Panel on a slide with about 15 mm of travel around 280 mm; mirror and panel fixed to well under 0.5 mm once set.
3. Measure f with a distant lamp; set the panel by parallax against a target at 4.9 m after the rig has warmed up; record the result for the renderer.
4. Calibrate as the plan describes (eye point, virtual-plane projection, pre-distortion mesh), adding a per-colour mesh and storing the measured image distance across the field, not one plane (sections 4 and 6).
5. Inside the box: matte black everywhere, light only the crop, keep the rest of the panel black (section 8).
6. Combiner: measure its reflection and the back-surface ghost with the panel's own polarisation, in both orientations (section 9).
7. Sun: lid closed whenever the HUD is off; never leave the rig in a parked car in the sun; sun-watch rule and thermistor as in Stage 2; thermocouple test of the empty box before daytime use (section 10).
8. Cleaning: water and a soft brush or microfibre on the grooved face; no ammonia or alcohol cleaners on PMMA.

A Appendix: how the numbers were made

A.1 Inputs

QUANTITY	VALUE	SOURCE
Focal length f, panel distance u	300 mm, 280 mm	Plan, `calc.py` F1 and U1 (Stage 1)
Crop	80 × 40 mm	Plan, section 5
Lens size, thickness, groove pitch, material	300 × 300 mm, 2.0 mm, 0.5 mm, PMMA	Knight Optical LFQ300300 stock listing
Focal-length tolerance	±2% precision, ±5% standard	Knight Optical (Photonics.com listing, 2017)
PMMA refractive index	1.491 at 587.6 nm; 1.4998 / 1.4935 / 1.4893 at 450 / 530 / 620 nm	Szczurowski (2012) dispersion formula
PMMA expansion and dn/dT	70 ppm/K, -1.05×10^{-4}/K	Typical values; approximate
Eye to lens along the light path	700 mm (550 + 150)	ASSUMED
Design conjugate of the stock lens	parallel light on the grooved side	ASSUMED (standard for stock lenses)
Windshield and combiner transmission	70% each	Plan (windshield ASSUMED there); combiner 70/30
Groove and draft loss near the centre	5%	ASSUMED; the trace gives 0–14% across the field

A.2 The lens model and the ray trace

The sheet is modelled with its flat face at z = –T and its grooved face collapsed onto z = 0: at each radius the grooved face has the slope of the facet there but no height. For each radius from 0 to 230 mm (0.05 mm steps), the facet slope is found by bisection so that a ray parallel to the axis, arriving on the grooved side, refracts through both surfaces and crosses the axis at the back focal point, f – T/n beyond the flat face. Integrating the slopes gives the solid asphere of Figure 4. All surfaces are traced exactly with the vector form of Snell's law; Fresnel transmission is computed for both polarisations at every surface.

Stage 1 is then traced backward from the eye: a ray leaves the eye point in a chosen direction, refracts at the grooved face (using the facet normal at the radius it hits), crosses the sheet, refracts at the flat face and lands on the panel. This gives the map from viewing direction to panel point, which is the distortion of section 6 and the field of view. Image distance at a field point comes from launching two rays from the panel point, a hair apart in the tangential and then the sagittal direction, tracing them forward and finding where their extensions cross. The on-axis result reproduces the thin-lens value, –4,200 mm, to better than 0.1 mm, which checks the facet solution, the principal-plane offset and the tracer together.

A.3 The groove-level trace

For sections 5, 7 and 8 the real profile is built in a plane through the axis: in each 0.5 mm zone the facet runs from its tip down to the valley, following the asphere with 20 points, and a vertical draft wall rises to the next tip. Rays are followed through every surface they meet, refracting or, past the critical angle, reflecting, until they leave the sheet. Partial Fresnel reflections are not followed (the Cycles renders include them). Sight lines from the eye give the shares in Figure 28; rays from the panel's centre give the orientation comparison of Figure 27. With straight (conical) facets the same trace gives the image steps of section 5.

A.4 The Cycles renders

The lens is built in Blender as real geometry: every tooth out to r = 160 mm, 1,024 segments around, 300+ zones, about 1.3 million vertices, with exact asphere normals on each facet (or plain flat facets for the comparison). The material is Cycles' glass shader, which includes surface reflection, at PMMA's index for each colour. The panel is an emitting rectangle showing the HUD test frame at 280 mm from the lens's principal plane, inside a matte black box. Each picture merges three renders made at 620, 530 and 450 nm, each emitting only that colour of the test frame. The "eye" camera has a 4 mm aperture focused at 4.9 m; the pinhole camera has none. Ghost levels were measured on the linear render data. Scripts: `blender/through_lens.py`, `blender/measure_ghosts.py`.

A.5 Formulas

Lens equation (u from H, v from H′):	$1/v = 1/f - 1/u \qquad v = -uf / (f - u)$
Magnification:	$M = \lvert v \rvert / u = f / (f - u)$
Sensitivity:	$dv/du = M^2$ in dioptres: $d(1/v)/du = 1/u^2$
Parallax error for head offset e:	$e \times \lvert 1/D_{true} - 1/D_{assumed} \rvert$ (radians)
Pupil footprint on the lens:	pupil × D_image / (D_image + eye distance)
Diffraction across a groove (FWHM):	$0.886\ \lambda$ / pitch
Flat-facet image step:	≈ pitch / f (traced: 5.1′)
Combiner ghost offset:	$2\ t \tan \theta_t \cos \theta_i$, with $\sin \theta_t = \sin \theta_i / n$
Luminance at the eye:	L_panel × T_lens × (1 − groove loss) × R_combiner

A.6 Glossary

Abbe number. A material's resistance to splitting colours; higher means less colour fringing.

Astigmatism. Lines in two directions coming into focus at different distances.

Combiner. A partly reflective glass sheet that shows the HUD image and the road together.

Critical angle. The angle inside a material beyond which light cannot leave and reflects completely; 42.1° for PMMA.

Dioptre (D). One over a distance in metres; the unit the eye focuses in.

Draft wall. The near-vertical step between two teeth of a Fresnel lens.

Eye box. The region the eye can move in and still see the whole image.

Facet. The slanted face of a tooth, which does the bending.

Field curvature. An image that lies on a curved surface instead of a plane.

Focal length. The distance at which a lens brings parallel light to a point.

Ghost. A faint extra copy of an image from an unwanted reflection.

Lateral colour. Red, green and blue images of different sizes, so edges show colour fringes toward the edge of the field.

Parallax. Two objects at different distances shifting against each other when the viewpoint moves.

Pitch. The spacing of the grooves.

PMMA. Poly(methyl methacrylate), acrylic: the clear plastic most Fresnel lenses are made of.

Principal plane. The reference plane from which a thick lens's distances are measured.

s and p polarisation. Light vibrating parallel to a surface (s) or in the plane of incidence (p); surfaces reflect s more.

Virtual image. The place light appears to come from after a lens or mirror, where no light actually is.

Sources

- Windshield AR HUD plan, TFT LCD edition (`docs/ARHUD-PLAN-TFT.pdf`), sections 5, 6.8, 7.5 and the gates; numbers from `docs/arhud-plan-tft-src/calc.py`.
- Knight Optical, Fresnel lenses, stock listing (LFQ300300: 300 mm focal length, 300 × 300 mm, 2 mm thick, 0.5 mm facet width, precision grade), knightoptical.com/stock/lenses/fresnel-lenses; capabilities sheet (PMMA standard, glass to order).
- Photonics.com product listing for Knight Optical Fresnel lenses (2017): focal-length tolerance ±2% precision, ±5% standard; AR coatings available.
- Edmund Optics, Fresnel lenses: acrylic, n = 1.49, about 92% transmission 400–1100 nm; grooved side toward the longer conjugate; total internal reflection limits the aperture near f/1 with the flat side toward the longer conjugate; high distortion. edmundoptics.com/c/fresnel-lenses/730/
- N. Sultanova et al. (2009) and M. Szczurowski (2012), PMMA and polycarbonate dispersion, via refractiveindex.info; Schott BK7.
- SpecialChem, maximum continuous service temperature (PMMA 65–95 °C); MakeItFrom, PMMA properties (heat deflection 96 °C); US 10,043,930 B2 (acrylic service temperature, dn/dT of PMMA).
- Texas Instruments DLPA096 and DLPA057 (production HUD sun load and virtual-image distance), as cited in the plan.